

Chapter 6: Geometrical figures

10 lessons · one lesson for each exercise · each lesson is one hour

NSW syllabus outcome

Stage 4 Measurement and space: MA4-ANG-C-01 (angle relationships, triangles and quadrilaterals) and transformations of shapes on the Cartesian plane and grids.



6.01 Transforming shapes

6.02 Combined transformations

6.03 Line symmetry

6.04 Rotational symmetry

6.05 Classifying triangles

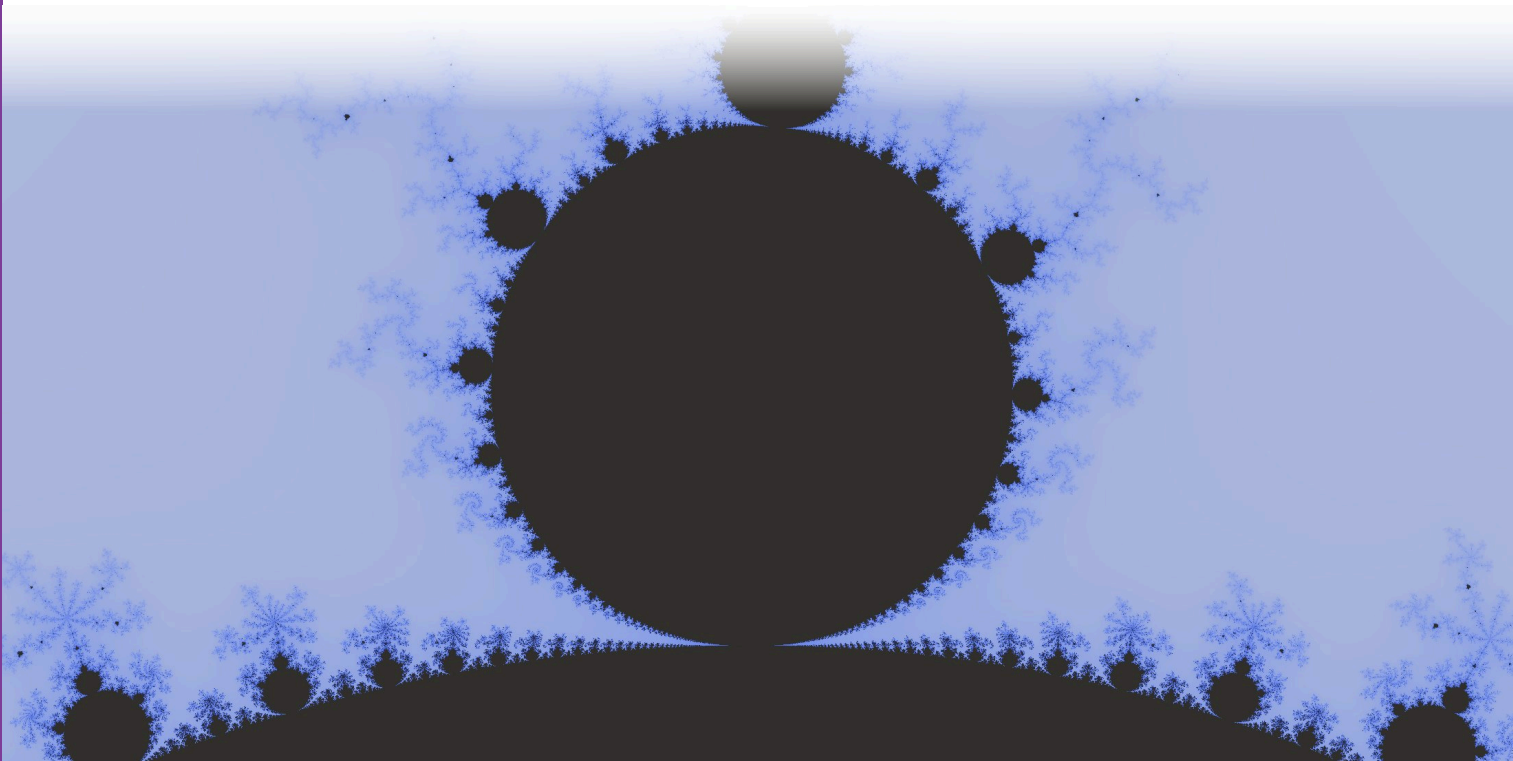
6.06 Angle sum of a triangle

6.07 Exterior angle of a triangle

6.08 Quadrilaterals

6.09 Angle sum of a quadrilateral

6.10 Properties of quadrilaterals



How to use this booklet

1 Each lesson is **one hour**. Start with the **learning intentions** and **success criteria**, write the **key terms** and use the **notes** box.

WE DO **Grey boxes.** Your teacher works each example (parts a, b, c) on the board. **Copy every step** into the space.

2 Every lesson has three levels: **Easy**, then **Medium**, then **Challenging**. Work down the page: the questions get harder as you go.

YOU DO **Blue boxes.** Now it is your turn. Work on your own. Quick drill answers go on the line; show working in the boxes.

DRILL **Skill drill pages.** After the Easy and the Medium examples there is a full page of short practice questions. Work down the page, write answers on the lines, then mark it and record your score and time.

3 Finished early? Try the **extension**. At the end, **show off your skill** on the tear-off exit ticket and hand it in.

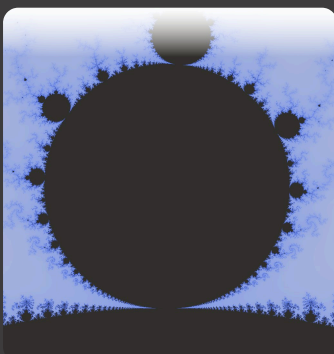
4 Missed a lesson? Each lesson ends with a **summary** with a worked example for every level. Start there.

Contents and progress tracker

Lesson	Topic	Page	Date	Done
6.01	Transforming shapes	3		☐
6.02	Combined transformations	11		☐
6.03	Line symmetry	19		☐
6.04	Rotational symmetry	27		☐
6.05	Classifying triangles	35		☐
6.06	Angle sum of a triangle	43		☐
6.07	Exterior angle of a triangle	51		☐
6.08	Quadrilaterals	59		☐
6.09	Angle sum of a quadrilateral	67		☐
6.10	Properties of quadrilaterals	75		☐

In this chapter you will

- reflect, translate and rotate shapes
- find line and rotational symmetry
- classify triangles and quadrilaterals
- use the angle sums of triangles and quadrilaterals
- use the exterior angle of a triangle
- describe the properties of special quadrilaterals



About the artwork

The pictures in this booklet are parts of the **Mandelbrot set**, drawn by a computer that repeats one simple rule, $z \rightarrow z^2 + c$, thousands of times for every point. Every booklet shows a different part of the set: this one is the top bulb of the set and the lacy edge around it. Maths can make beautiful things.

Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

LESSON
6.01

Transforming shapes

Date _____

Learning intentions

We are learning to:

- reflect, translate and rotate a shape on a grid

Success criteria

I can:

- ☐ **reflect** a shape across a mirror line
- ☐ **translate** a shape a given distance left, right, up or down
- ☐ **rotate** a shape 90° , 180° or 270° about a point

Key terms Write the definition as your teacher explains it.

Reflect

Translate

Rotate

Notes

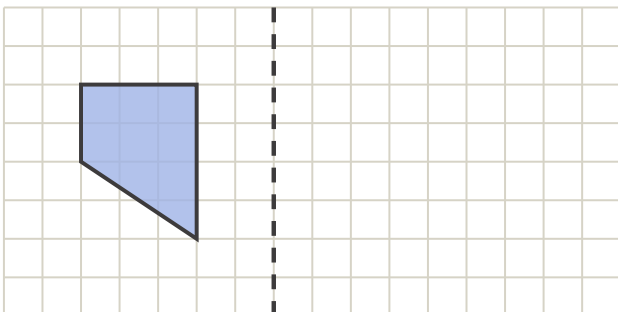
Level 1 · Getting started

Easy · 10 questions

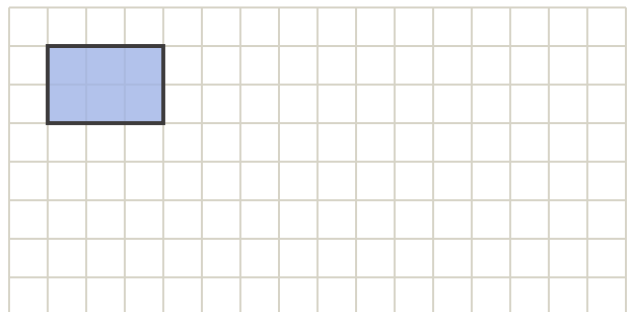
WE DO With your teacher: copy what they write on the board.

Example 1 Reflect, then translate the shape.

a Reflect the shape in the dotted line.



b Translate the shape 5 right and 3 down.

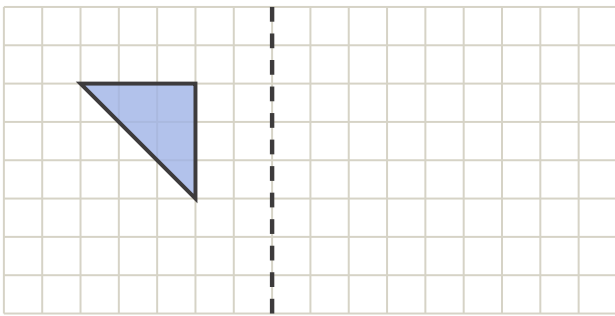


YOU DO On your own: **Easy** Set A. Reflect each shape in the dotted line.

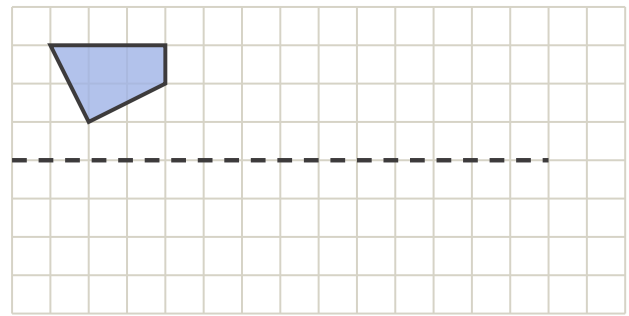
2 QUESTIONS

REMEMBER REFLECT: flip over the mirror line; same distance each side.

1



2



YOU DO On your own: **Easy** Set A2. Quick drill: a point is at (3, 2) on a grid. Where is it after the translation?

6 QUESTIONS

1 2 right

2 3 up

3 1 left, 2 down

4 4 right, 1 up

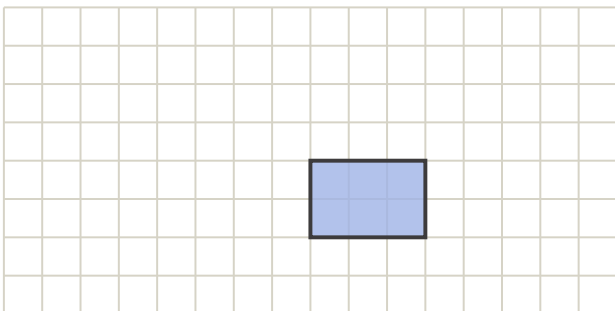
5 3 left, 2 up

6 5 right, 2 down

YOU DO On your own: **Easy** Set B. Translate each shape.

2 QUESTIONS

1 4 units left, 1 unit up



2 4 units right, 6 units up



SKILL DRILL 1 Easy basics: practise until it is automatic

40 questions

YOU DO On your own: **Easy** Skill drill 1. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 40 Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–24: translate (slide) the point. Write where it ends up.

1 $(-1, -6)$: 1 left, 3 down

2 $(2, -5)$: 1 right, 2 down

3 $(6, -2)$: 4 right

4 $(-3, -4)$: 3 right

5 $(2, 0)$: 6 right, 6 down

6 $(4, -2)$: 5 left, 1 up

7 $(3, 5)$: 1 left, 2 up

8 $(0, 3)$: 1 left, 2 up

9 $(-3, 4)$: 3 left, 4 down

10 $(4, 2)$: 4 left, 1 up

11 $(3, 2)$: 2 left, 3 down

12 $(1, -4)$: 4 left, 1 up

13 $(-4, -4)$: 6 right, 6 up

14 $(-4, -5)$: 5 left, 2 down

15 $(-2, -1)$: 1 right, 6 down

16 $(2, -3)$: 1 left, 4 down

17 $(-6, -1)$: 6 right

18 $(-4, -3)$: 3 left, 1 down

19 $(-2, 3)$: 5 left, 5 down

20 $(-5, -5)$: 1 right, 3 down

21 $(4, 0)$: 2 left, 2 down

22 $(2, 0)$: 4 left, 5 up

23 $(2, -4)$: 6 left, 2 down

24 $(-4, -5)$: 2 left, 2 down

ROUND 2 Questions 25–40: reflect (flip) the point in the axis given.

25 $(-2, 6)$ in the x-axis

26 $(-2, 1)$ in the x-axis

27 $(-6, 2)$ in the x-axis

28 $(8, 6)$ in the x-axis

29 $(2, -3)$ in the x-axis

30 $(-7, -6)$ in the y-axis

31 $(-8, -4)$ in the y-axis

32 $(-3, 1)$ in the x-axis

33 $(-5, 8)$ in the x-axis

34 $(-4, 1)$ in the x-axis

35 $(3, -2)$ in the x-axis

36 $(-1, -2)$ in the x-axis

37 $(-6, 2)$ in the y-axis

38 $(6, -6)$ in the x-axis

39 $(-7, 4)$ in the y-axis

40 $(6, -4)$ in the x-axis

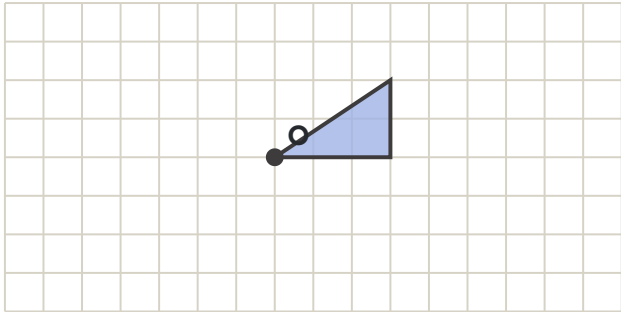
Level 2 · Building up

Medium · 2 questions

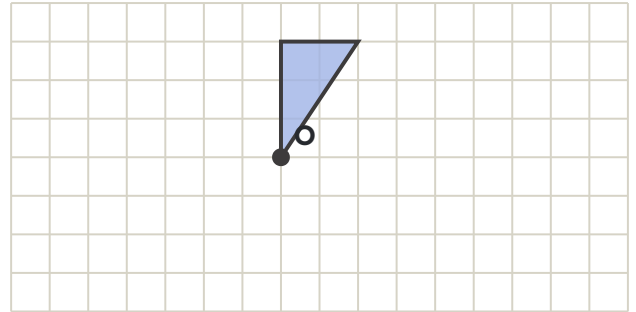
WE DO With your teacher: copy what they write on the board.

Example 2 Rotate the shape about O.

a Rotate 180° about O.



b Rotate 90° clockwise about O.

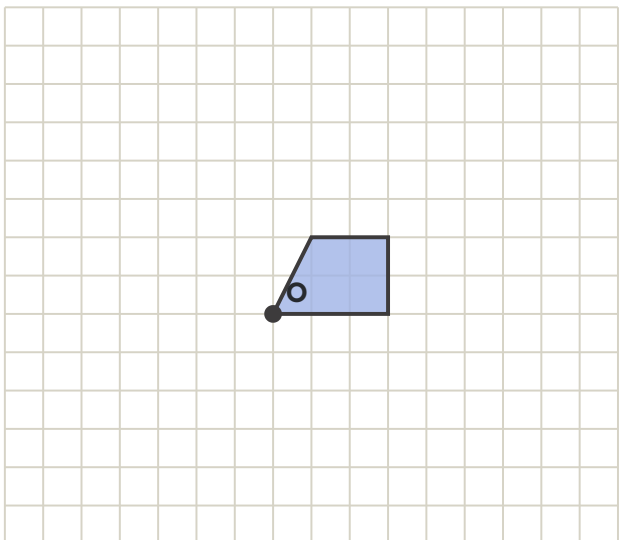


YOU DO On your own: **Medium** Set C. Rotate each shape about O.

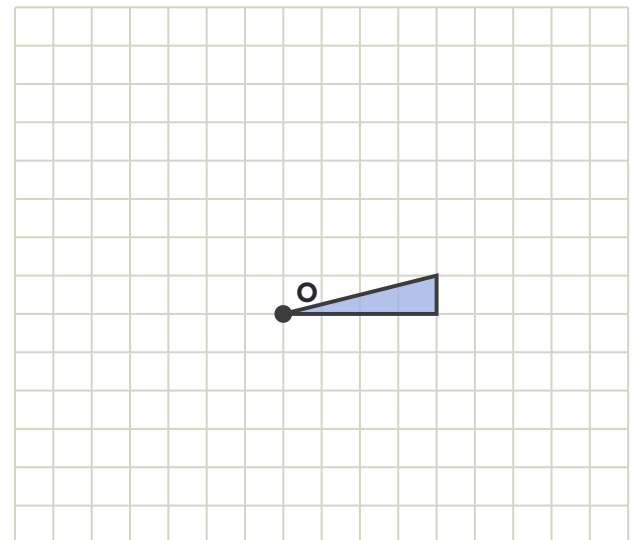
2 QUESTIONS

REMEMBER TRANSLATE: slide; count units left/right then up/down.

1 180°



2 90° clockwise



SKILL DRILL 2 Medium basics: keep practising

30 questions

YOU DO

On your own: **Medium** Skill drill 2. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 30

Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–15: rotate (turn) the point about the origin.

1 (1, 4), 90° clockwise

2 (–2, –5), 90° clockwise

3 (–2, 6), 180°

4 (4, 5), 180°

5 (6, 6), 90° anticlockwise

6 (–3, –4), 90° clockwise

7 (–5, 5), 90° clockwise

8 (–6, –3), 180°

9 (–4, –4), 90° clockwise

10 (2, 4), 90° clockwise

11 (4, 1), 180°

12 (–6, –4), 180°

13 (5, –2), 90° clockwise

14 (–6, –1), 180°

15 (–3, –6), 180°

ROUND 2 Questions 16–30: name the transformation that moves the first point to the second.

16 (4, 5) → (7, 5)

17 (–1, 5) → (1, 5)

18 (1, 7) → (–1, –7)

19 (4, 1) → (–4, –1)

20 (7, –6) → (–7, 6)

21 (–4, 7) → (–1, 7)

22 (–7, 4) → (–7, –4)

23 (6, –1) → (9, –1)

24 (4, –7) → (7, –7)

25 (7, 2) → (–7, 2)

26 (1, –6) → (–1, –6)

27 (–4, 6) → (4, 6)

28 (–7, –1) → (–7, 1)

29 (–5, –6) → (–5, 6)

30 (6, 1) → (–6, 1)

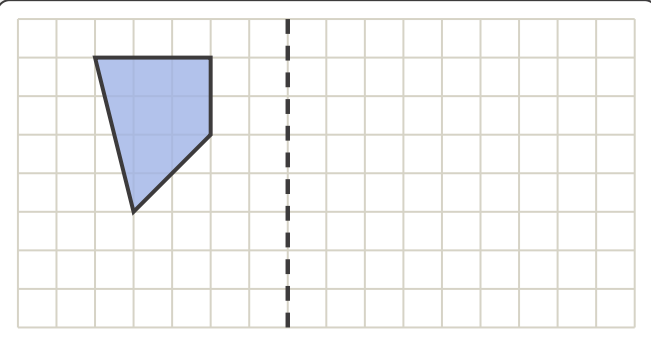
Level 3 · Challenge

Challenging · 2 questions

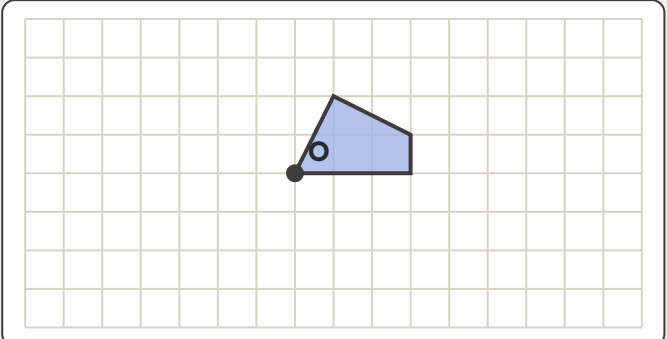
WE DO With your teacher: copy what they write on the board.

Example 3 Reflect, then rotate the shape.

a Reflect in the dotted line.



b Rotate 270° clockwise about O.

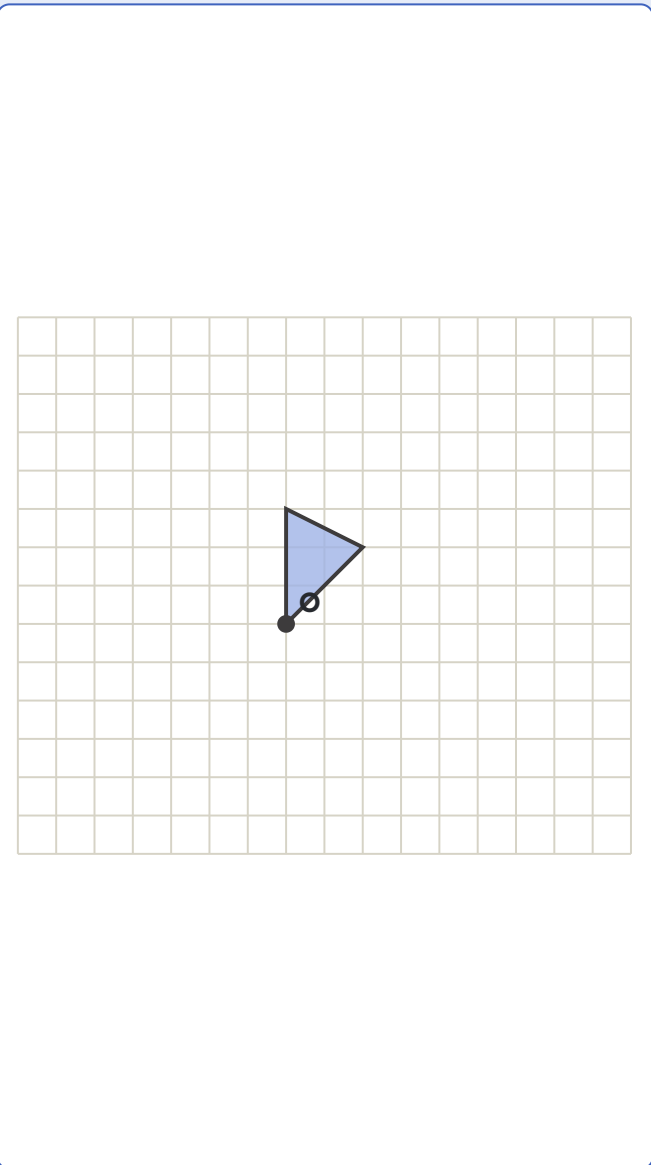


YOU DO On your own: **Challenging** Set D. Transform each shape.

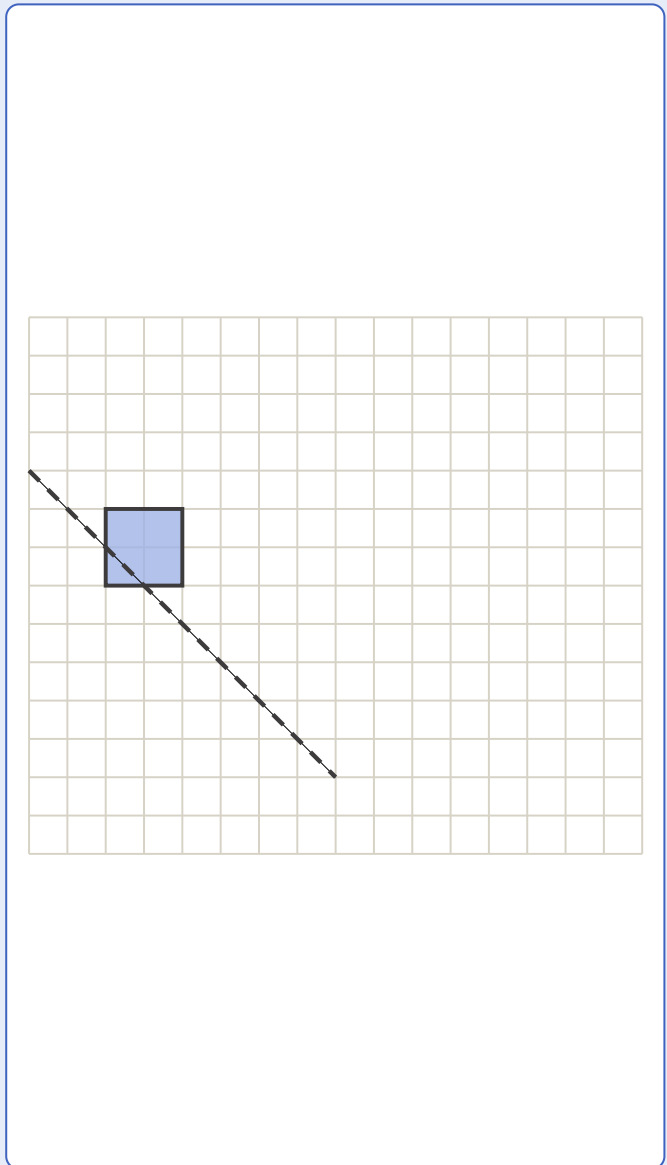
2 QUESTIONS

REMEMBER **ROTATE**: turn about a point; use tracing paper.

1 Rotate 270° clockwise about O.



2 Reflect in the dotted line.





Extension · Stretch yourself

Finished Set D? Stretch yourself here.



Worked example

A shape is rotated 90° clockwise and then 180° .
What single rotation does the same thing?

1. Rotations about the same point add
2. $90^\circ + 180^\circ = 270^\circ$

Answer: 270° clockwise (or 90° anticlockwise)

YOU DO

On your own: read the worked example,
then try these.

- E1 Which single transformation is the same as reflecting in a vertical line and then in a horizontal line that crosses it?

- E2 Is a 90° clockwise rotation the same as a 270° anticlockwise rotation? Explain.

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO

Show off your skill

Choose **ONE** question. Show
your working.

Name _____ Class _____

Level 1

Name the transformation that flips
a shape.

Level 2

A point is at (2, 3) on a grid. Where
is it after a translation 4 right and 1
down?

Level 3

Describe what a 90° clockwise
rotation does to an arrow pointing
up.

Lesson summary

Missed the lesson? Start here.

- Remember**
1. **Reflect:** flip over the mirror line; same distance each side.
 2. **Translate:** slide; count units left/right then up/down.
 3. **Rotate:** turn about a point; use tracing paper.
 4. The shape keeps its size and shape.

Level 1

Reflect in a vertical line

1. Each vertex the same distance on the other side

Answer: a mirror image

Level 2

Translate 5 right, 3 down

1. Move every vertex

Answer: same shape, new position

Level 3

Rotate 180° about O

1. Each vertex goes through O to the other side

Answer: upside down

Extension

90° then 180°

1. Add

Answer: 270° clockwise

My reflection: what did I learn today? Which question was hardest, and why?

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
6.02

Combined transformations

Date _____

Learning intentions

We are learning to:

- apply two or more transformations in order
- describe a combined transformation

Success criteria

I can:

- ☐ do each transformation in the order given
- ☐ label each new image (A, A', A'')
- ☐ describe a combination that maps one shape onto another

Key terms Write the definition as your teacher explains it.

Image

Combined transformation

Notes

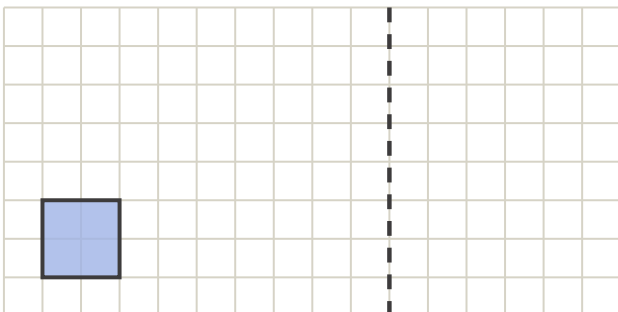
Level 1 · Getting started

Easy · 10 questions

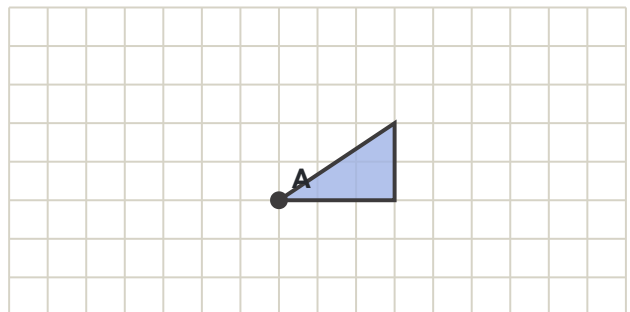
WE DO With your teacher: copy what they write on the board.

Example 1 Do both transformations in order.

a Translate 5 right, 2 up; then reflect.



b Rotate 90° clockwise about A, then 3 left.

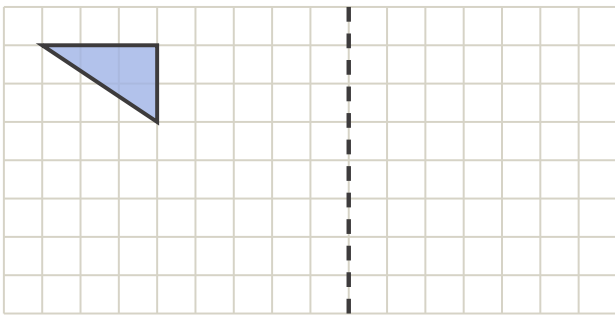


YOU DO On your own: **Easy** Set A. Do both transformations.

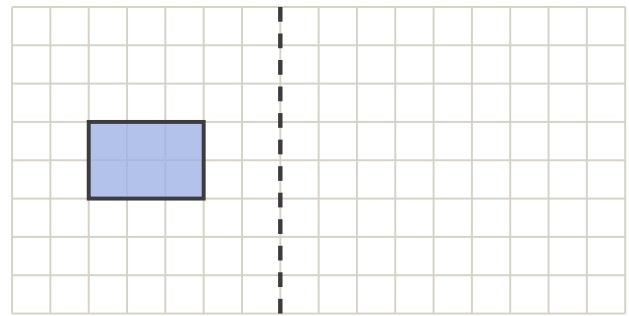
2 QUESTIONS

REMEMBER Do the transformations in the order given.

1 Translate 3 right; then reflect.



2 Reflect; then translate 2 up.



YOU DO On your own: **Easy** Set A2. Quick drill: name the single transformation.

6 QUESTIONS

1 a flip

2 a slide

3 a turn

4 a mirror image

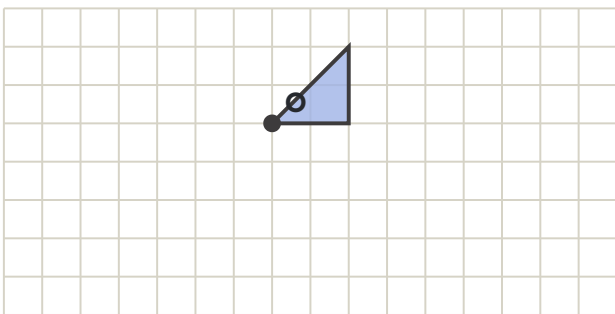
5 moves every point 3 right

6 turns about a point

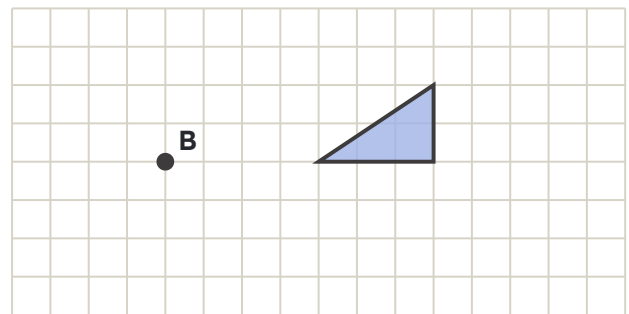
YOU DO On your own: **Easy** Set B. Do both transformations.

2 QUESTIONS

1 Rotate 180° about O, then translate 2 down.



2 Translate 4 left, then rotate 90° clockwise about B.



SKILL DRILL 1 Easy basics: practise until it is automatic

29 questions

YOU DO

On your own: **Easy** Skill drill 1. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 29

Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–14: name the transformation: translation, reflection or rotation.

1 a flip

2 a butterfly's two wings

3 moving 5 units up

4 a clock hand moving

5 a door opening

6 a turn

7 a wheel turning

8 a mirror image

9 turning 90° about a point

10 a car driving straight

11 your reflection in a lake

12 a slide

13 a chess rook moving

14 a lift going up

ROUND 2 Questions 15–29: two translations. Write the single translation that does the same.

15 7 right, then 5 left

16 7 left, then 5 left

17 6 right, then 4 right

18 1 right, then 3 right

19 7 right, then 7 right

20 6 left, then 2 right

21 5 right, then 3 left

22 7 right, then 1 left

23 7 right, then 2 left

24 6 right, then 5 right

25 1 left, then 7 left

26 4 left, then 3 left

27 4 right, then 3 left

28 4 left, then 2 left

29 5 left, then 7 left

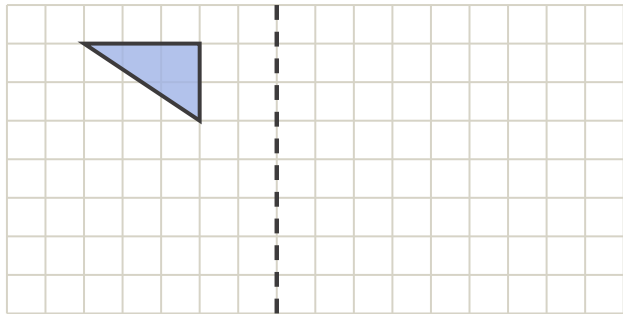
Level 2 · Building up

Medium · 2 questions

WE DO With your teacher: copy what they write on the board.

Example 2 Do both transformations, then compare.

a Reflect in the dotted line, then translate 1 down.



b Does the order matter? Try reflecting then translating versus translating then reflecting.

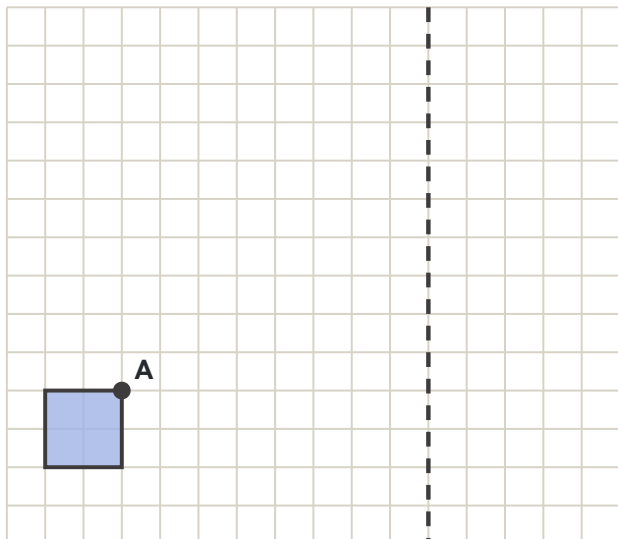
Your teacher works this out. Copy every step.

YOU DO On your own: **Medium** Set C. Do all three transformations.

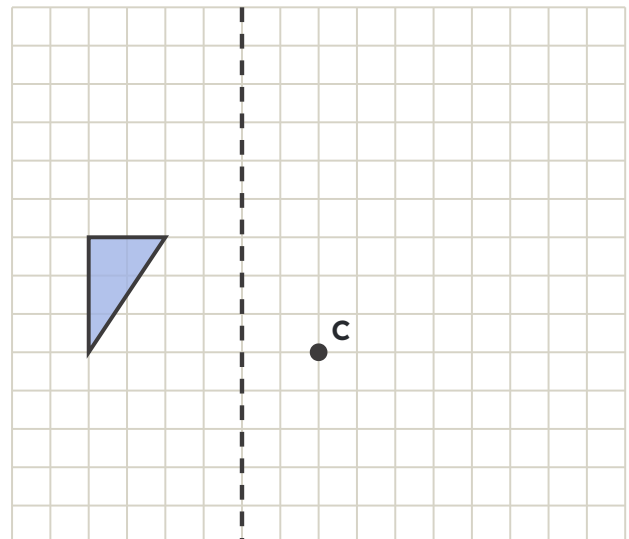
2 QUESTIONS

REMEMBER Draw each image before doing the next.

1 Translate 5 right, 2 up; rotate 90° clockwise about A; reflect in the dotted line.



2 Reflect in the dotted line; translate 1 left; rotate 180° about C.



SKILL DRILL 2 Medium basics: keep practising

24 questions

YOU DO

On your own: **Medium** Skill drill 2. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 24 Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–12: do both transformations in order. Where does the point end up?

1 $(-5, -5)$: 1 right, then reflect in the y-axis

2 $(4, 3)$: 4 right, then reflect in the x-axis

3 $(-5, 5)$: 1 right, then reflect in the y-axis

4 $(1, 4)$: 1 right, then reflect in the y-axis

5 $(-1, 1)$: 5 right, then reflect in the x-axis

6 $(-2, 5)$: 4 right, then reflect in the y-axis

7 $(2, 5)$: 5 right, then reflect in the x-axis

8 $(-1, 5)$: 5 right, then reflect in the y-axis

9 $(-5, 5)$: 5 right, then reflect in the y-axis

10 $(-3, 4)$: 1 right, then reflect in the x-axis

11 $(3, -2)$: 2 right, then reflect in the y-axis

12 $(2, 3)$: 3 right, then reflect in the y-axis

ROUND 2 Questions 13–24: the point moved from the first position to the second. Describe the single translation.

13 $(-3, -6) \rightarrow (-9, -7)$

14 $(4, 2) \rightarrow (3, -1)$

15 $(4, -3) \rightarrow (9, 0)$

16 $(5, 1) \rightarrow (11, 0)$

17 $(3, -3) \rightarrow (4, -2)$

18 $(-6, 6) \rightarrow (-1, 10)$

19 $(4, 3) \rightarrow (0, 6)$

20 $(-5, -4) \rightarrow (-7, -6)$

21 $(0, -2) \rightarrow (1, -1)$

22 $(-2, 5) \rightarrow (2, 10)$

23 $(6, -1) \rightarrow (1, -5)$

24 $(4, 6) \rightarrow (5, 11)$

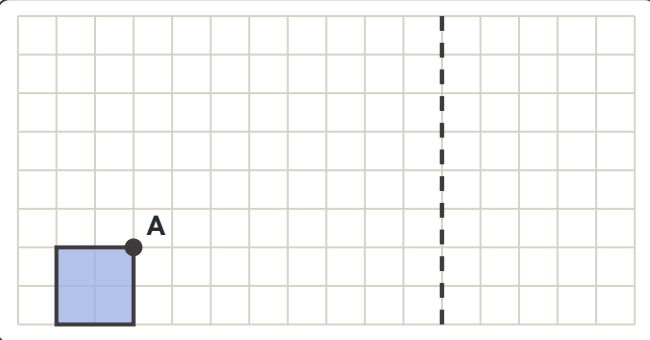
Level 3 · Challenge

Challenging · 4 questions

WE DO With your teacher: copy what they write on the board.

Example 3 Do all three transformations, then describe a combination.

- a** Translate 5 right and 2 up, rotate 90° clockwise about A, then reflect in the dotted line.



- b** Describe two transformations that move a shape from the top-left to the bottom-right of a grid, upside down.

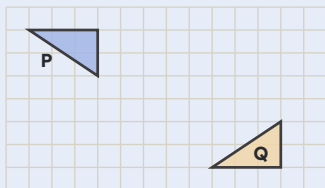
Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Describe the transformations.

4 QUESTIONS

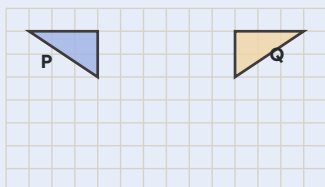
REMEMBER Label them A, A', A''.

- 1** Describe how shape P moves onto shape Q.



- 3** Does a translation followed by a reflection give the same result as the reflection first? Give an example.

- 2** Find two different ways to move P onto Q.



- 4** Which transformations never change the direction a shape faces?



Extension · Stretch yourself

Finished Set D? Stretch yourself here.



Worked example

A triangle has vertices $(1, 1)$, $(3, 1)$, $(3, 4)$. It is reflected in the y -axis, then translated 2 units down. Where are the vertices?

1. Reflect in y -axis: $(-1, 1)$, $(-3, 1)$, $(-3, 4)$
2. Down 2: subtract 2 from each y

Answer: $(-1, -1)$, $(-3, -1)$, $(-3, 2)$

YOU DO

On your own: read the worked example, then try these.

- E1 Reflect $(2, 5)$ in the x -axis, then translate 3 left. Where is it?

- E2 Rotate $(4, 1)$ by 180° about the origin. Where is it?

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO

Show off your skill

Choose ONE question. Show your working.

Name _____ Class _____

Level 1

What does A' mean?

Level 2

A point $(1, 2)$ is translated 3 right then 4 up. Where is it?

Level 3

A point $(2, 3)$ is reflected in the x -axis then translated 2 left. Where is it?

Lesson summary

Missed the lesson? Start here.

- Remember**
1. Do the transformations in the order given.
 2. Draw each image before doing the next.
 3. Label them A, A', A''.
 4. Order can change the final position.

Level 1

Translate then reflect

1. Slide first
2. Then flip over the line

Answer: image A''

Level 2

Rotate then translate

1. Turn about the point
2. Then slide

Answer: image A''

Level 3

Order matters?

1. Try both ways

Answer: often yes

Extension

Reflect in y-axis then down 2

1. $(x, y) \rightarrow (-x, y - 2)$

Answer: $(-1, -1)$, $(-3, -1)$, $(-3, 2)$

My reflection: what did I learn today? Which question was hardest, and why?

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
6.03

Line symmetry

Date _____

Learning intentions

We are learning to:

- find the axes of symmetry of a shape

Success criteria

I can:

- ☐ know that an axis of symmetry divides a shape into two mirror images
- ☐ draw every axis of symmetry on a shape
- ☐ count the axes of symmetry

Key terms Write the definition as your teacher explains it.

Line symmetry

Axis of symmetry

Notes

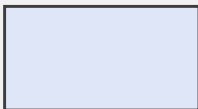
Level 1 · Getting started

Easy · 18 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Draw every axis of symmetry.

a Draw the axes of symmetry.



b Draw the axes of symmetry.



Your teacher works this out. Copy every step.

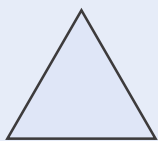
Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Draw all the axes of symmetry. Write how many.

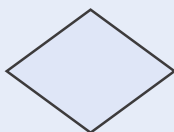
6 QUESTIONS

REMEMBER An **AXIS OF SYMMETRY** folds a shape onto itself.

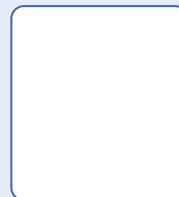
1



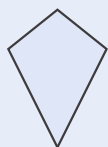
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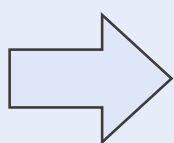
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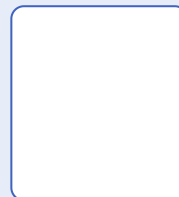
2



4



6



YOU DO On your own: **Easy** Set A2. Quick drill: how many axes of symmetry?

9 QUESTIONS

1 square

2 rectangle

3 equilateral triangle

4 isosceles triangle

5 regular hexagon

6 parallelogram

7 kite

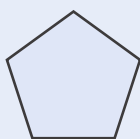
8 circle

9 regular pentagon

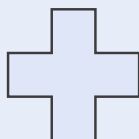
YOU DO On your own: **Easy** Set B. Draw all the axes of symmetry. Write how many.

3 QUESTIONS

1



2



3



SKILL DRILL 1 Easy basics: practise until it is automatic

34 questions

YOU DO

On your own: **Easy** Skill drill 1. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 34

Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–14: how many axes (lines) of symmetry?

1 kite

2 parallelogram

3 isosceles trapezium

4 isosceles triangle

5 regular hexagon

6 semicircle

7 scalene triangle

8 square

9 equilateral triangle

10 regular pentagon

11 regular octagon

12 rectangle

13 rhombus

14 circle

ROUND 2 Questions 15–34: how many axes of symmetry does each capital letter have?

15 S

16 M

17 O

18 F

19 H

20 D

21 A

22 T

23 E

24 X

25 N

26 K

27 L

28 C

29 B

30 W

31 I

32 V

33 Z

34 U

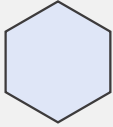
Level 2 · Building up

Medium · 6 questions

WE DO With your teacher: copy what they write on the board.

Example 2 How many axes of symmetry?

a How many axes?



b How many axes?



Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Medium** Set C. Draw all the axes of symmetry. Write how many.

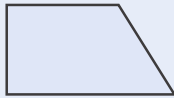
6 QUESTIONS

REMEMBER Test by folding or with a mirror.

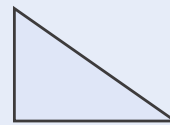
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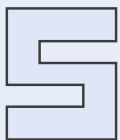
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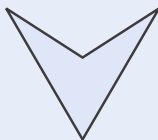
5



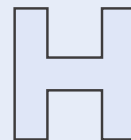
2



4



6



SKILL DRILL 2 Medium basics: keep practising

28 questions

YOU DO

On your own: **Medium** Skill drill 2. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 28 Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–16: how many axes of symmetry does a regular polygon with this many sides have?

- | | | | |
|---------------------|----------------------|----------------------|----------------------|
| 1 14 sides
_____ | 2 6 sides
_____ | 3 12 sides
_____ | 4 20 sides
_____ |
| 5 16 sides
_____ | 6 17 sides
_____ | 7 8 sides
_____ | 8 13 sides
_____ |
| 9 3 sides
_____ | 10 4 sides
_____ | 11 19 sides
_____ | 12 15 sides
_____ |
| 13 5 sides
_____ | 14 10 sides
_____ | 15 7 sides
_____ | 16 9 sides
_____ |

ROUND 2 Questions 17–28: true or false?

- | | |
|---|--|
| 17 A scalene triangle has no axes of symmetry.
_____ | 18 An isosceles triangle has 1 axis of symmetry.
_____ |
| 19 A rectangle's diagonals are axes of symmetry.
_____ | 20 A square has 4 axes of symmetry.
_____ |
| 21 A regular octagon has 4 axes of symmetry.
_____ | 22 A regular hexagon has 6 axes of symmetry.
_____ |
| 23 A kite has 1 axis of symmetry.
_____ | 24 The letter N has an axis of symmetry.
_____ |
| 25 A parallelogram has 2 axes of symmetry.
_____ | 26 A circle has infinitely many axes of symmetry.
_____ |
| 27 A rhombus's diagonals are axes of symmetry.
_____ | 28 Every triangle has an axis of symmetry.
_____ |

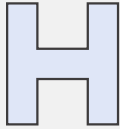
Level 3 · Challenge

Challenging · 4 questions

WE DO With your teacher: copy what they write on the board.

Example 3 How many axes of symmetry?

a How many axes?



Your teacher works this out. Copy every step.

b How many axes of symmetry does a circle have?

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Think it through.

4 QUESTIONS

REMEMBER Draw the axes as dotted lines.

1 How many axes of symmetry does a regular 10-sided shape have?

3 Draw a shape with exactly 2 axes of symmetry.

2 Which capital letters of the alphabet have exactly one axis of symmetry? List 5.

4 Can a triangle have exactly 2 axes of symmetry? Explain.

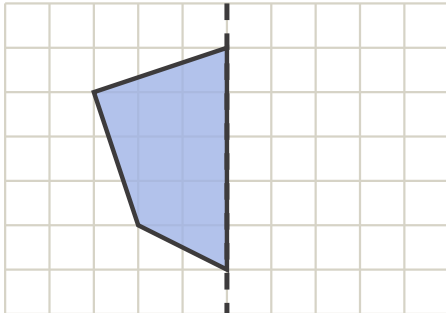
Extension · Stretch yourself

Finished Set D? Stretch yourself here.

Worked example

Complete the shape so that the dotted line is an axis of symmetry.

- Each point is reflected across the line
- Join the reflected points

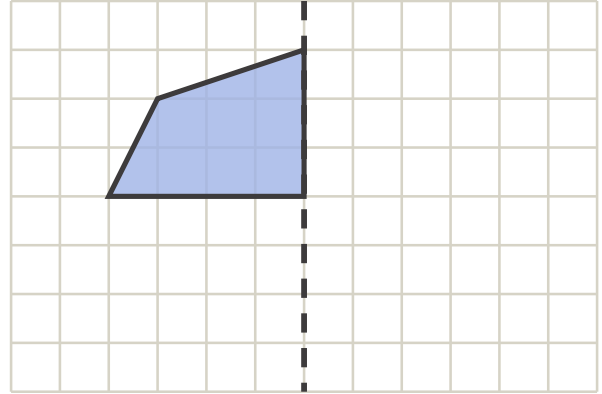


Answer: a symmetric shape

YOU DO

On your own: read the worked example, then try these.

- E1 Complete the shape so both dotted lines are axes of symmetry.



- E2 A regular polygon has 12 axes of symmetry. How many sides does it have?

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO

Show off your skill

Choose ONE question. Show your working.

Name _____ Class _____

Level 1

How many axes of symmetry does an equilateral triangle have?

Level 2

How many axes of symmetry does a kite have?

Level 3

How many axes of symmetry does a regular hexagon have?

Lesson summary

Missed the lesson? Start here.

- Remember**
1. An **axis of symmetry** folds a shape onto itself.
 2. Test by folding or with a mirror.
 3. Draw the axes as dotted lines.
 4. A regular polygon with n sides has n axes.

Level 1

Rectangle

1. Vertical and horizontal only

Answer: 2 axes

Level 2

Square

1. 2 through sides, 2 diagonals

Answer: 4 axes

Level 3

Parallelogram

1. No fold works

Answer: 0 axes

Extension

Regular 12-gon

1. n axes

Answer: 12 axes

My reflection: what did I learn today? Which question was hardest, and why?

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
6.04

Rotational symmetry

Date _____

Learning intentions

We are learning to:

- find the order of rotational symmetry of a shape

Success criteria

I can:

- ☐ know a shape has rotational symmetry if it looks the same before a full turn
- ☐ count the **order**: how many times it matches in one full turn
- ☐ find the angle of each turn

Key terms Write the definition as your teacher explains it.

Rotational symmetry

Order

Notes

Level 1 · Getting started

Easy · 18 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Give the order of rotational symmetry.

a Order of rotational symmetry?



b Order?



Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

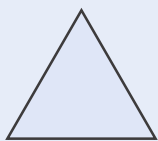
YOU DO

On your own: **Easy** Set A. Does the shape have rotational symmetry? If so, give its order.

6 QUESTIONS

REMEMBER Turn the shape about its centre.

1



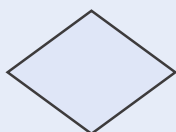
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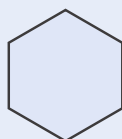
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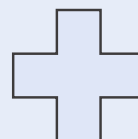
2



4



6



YOU DO

On your own: **Easy** Set A2. Quick drill: order of rotational symmetry (write 1 for none).

9 QUESTIONS

1 square

2 rectangle

3 equilateral triangle

4 kite

5 regular hexagon

6 parallelogram

7 letter S

8 letter A

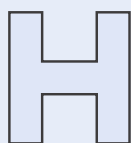
9 regular octagon

YOU DO

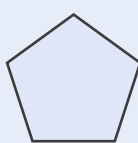
On your own: **Easy** Set B. Give the order of rotational symmetry.

3 QUESTIONS

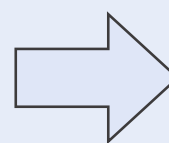
1



2



3



SKILL DRILL 1 Easy basics: practise until it is automatic

34 questions

YOU DO On your own: **Easy** Skill drill 1. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 34 Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–14: order of rotational symmetry (write 1 for none).

1 parallelogram

2 square

3 circle

4 rectangle

5 isosceles triangle

6 regular pentagon

7 kite

8 scalene triangle

9 rhombus

10 isosceles trapezium

11 regular octagon

12 equilateral triangle

13 semicircle

14 regular hexagon

ROUND 2 Questions 15–34: order of rotational symmetry of each capital letter (write 1 for none).

15 U

16 H

17 X

18 F

19 M

20 N

21 C

22 T

23 B

24 E

25 Z

26 V

27 S

28 K

29 I

30 D

31 W

32 O

33 A

34 L

Level 2 · Building up

Medium · 6 questions

WE DO With your teacher: copy what they write on the board.

Example 2 Give the order of rotational symmetry.

a Order?



Your teacher works this out. Copy every step.

b Order?



Your teacher works this out. Copy every step.

YOU DO On your own: **Medium** Set C. Give the order and the smallest angle of turn.

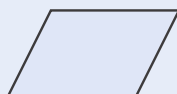
6 QUESTIONS

REMEMBER Count how many times it looks the same in one turn.

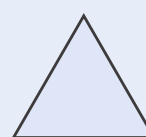
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3



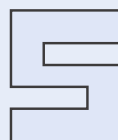
5



2



4



6



SKILL DRILL 2 Medium basics: keep practising

27 questions

YOU DO

On your own: **Medium** Skill drill 2. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 27 Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1

Questions 1–15: a regular polygon has this many sides. Write its order of rotational symmetry and the smallest angle of turn.

- | | | |
|----------------------|----------------------|----------------------|
| 1 9 sides
_____ | 2 3 sides
_____ | 3 36 sides
_____ |
| 4 10 sides
_____ | 5 40 sides
_____ | 6 18 sides
_____ |
| 7 6 sides
_____ | 8 8 sides
_____ | 9 24 sides
_____ |
| 10 20 sides
_____ | 11 15 sides
_____ | 12 45 sides
_____ |
| 13 5 sides
_____ | 14 4 sides
_____ | 15 12 sides
_____ |

ROUND 2 Questions 16–27: true or false?

- | | |
|---|--|
| 16 A regular hexagon has order 6.
_____ | 17 Every shape has rotational symmetry of at least order 1.
_____ |
| 18 A shape with order 2 looks the same upside down.
_____ | 19 A rectangle has rotational symmetry of order 4.
_____ |
| 20 A square has rotational symmetry of order 4.
_____ | 21 The letter S has rotational symmetry of order 2.
_____ |
| 22 An equilateral triangle turns onto itself every 120° .
_____ | 23 A regular pentagon turns onto itself every 60° .
_____ |
| 24 A rhombus has order 2.
_____ | 25 A parallelogram has rotational symmetry.
_____ |
| 26 The letter T has rotational symmetry of order 2.
_____ | 27 A kite has rotational symmetry.
_____ |

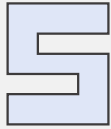
Level 3 · Challenge

Challenging · 4 questions

WE DO With your teacher: copy what they write on the board.

Example 3 Give the order, then find the angle.

a Order?



Your teacher works this out. Copy every step.

b A shape has order 5. What is the smallest angle it turns to match itself?

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Think it through.

4 QUESTIONS

REMEMBER That count is the **ORDER**.

1 Which capital letters have rotational symmetry of order 2? List 4.

3 A shape matches itself every 72° . What is its order?

2 Can a shape have rotational symmetry but no line symmetry? Give an example.

4 Draw a shape with rotational symmetry of order 4.



Extension · Stretch yourself

Finished Set D? Stretch yourself here.



Worked example

A regular polygon matches itself every 40° . How many sides does it have?

1. A full turn is 360°
2. $360 \div 40 = 9$

Answer: 9 sides

YOU DO

On your own: read the worked example, then try these.

- E1 A regular polygon matches itself every 30° . How many sides does it have?

- E2 Which has the larger angle of turn: order 6 or order 8? Explain.

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO

Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

What is the order of rotational symmetry of a square?

Level 2

Does a kite have rotational symmetry?

Level 3

A shape has order 6. Find its smallest angle of turn.

Lesson summary

Missed the lesson? Start here.

- Remember**
1. Turn the shape about its centre.
 2. Count how many times it looks the same in one turn.
 3. That count is the **order**.
 4. Order 1 means no rotational symmetry. Angle = $360^\circ \div \text{order}$.

Level 1

Rectangle

1. Matches at 180° and 360°

Answer: order 2

Level 2

Square

1. Every 90°

Answer: order 4

Level 3

Kite

1. Only at 360°

Answer: no rotational symmetry

Extension

Every 40°

1. $360 \div 40$

Answer: 9 sides

My reflection: what did I learn today? Which question was hardest, and why?

This space is left blank on purpose. It is the back of your exit ticket.

LESSON 6.05

Classifying triangles

Date _____

Learning intentions

We are learning to:

- classify triangles by their sides and by their angles

Success criteria

I can:

- ☐ name triangles by sides: equilateral, isosceles, scalene
- ☐ name triangles by angles: acute-angled, right-angled, obtuse-angled
- ☐ find equal angles in an isosceles triangle

Key terms Write the definition as your teacher explains it.

Isosceles

Scalene

Equilateral

Notes

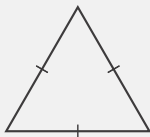
Level 1 · Getting started

Easy · 15 questions

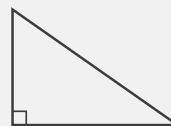
WE DO With your teacher: copy what they write on the board.

Example 1 Classify each triangle by its sides and by its angles.

a Classify by sides and by angles.



b Classify by sides and by angles.



Your teacher works this out. Copy every step.

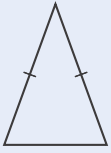
Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Classify each triangle by its sides.

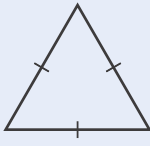
6 QUESTIONS

REMEMBER Sides: **EQUILATERAL** (3 equal), **ISOSCELES** (2 equal), **SCALENE** (none).

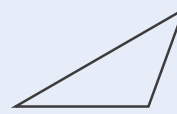
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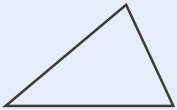
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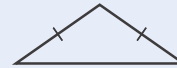
2



4



6



YOU DO On your own: **Easy** Set A2. Quick drill: classify by angles (acute, right or obtuse).

6 QUESTIONS

1 $60^\circ, 60^\circ, 60^\circ$

2 $90^\circ, 45^\circ, 45^\circ$

3 $120^\circ, 30^\circ, 30^\circ$

4 $50^\circ, 60^\circ, 70^\circ$

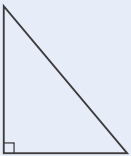
5 $100^\circ, 40^\circ, 40^\circ$

6 $30^\circ, 60^\circ, 90^\circ$

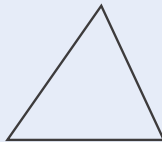
YOU DO On your own: **Easy** Set B. Classify each triangle by its angles.

3 QUESTIONS

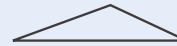
1



2



3



SKILL DRILL 1 Easy basics: practise until it is automatic

40 questions

YOU DO On your own: **Easy** Skill drill 1. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 40 Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–24: classify by angles: acute, right or obtuse.

1 $115^\circ, 42^\circ, 23^\circ$

2 $37^\circ, 86^\circ, 57^\circ$

3 $118^\circ, 33^\circ, 29^\circ$

4 $60^\circ, 76^\circ, 44^\circ$

5 $64^\circ, 29^\circ, 87^\circ$

6 $103^\circ, 35^\circ, 42^\circ$

7 $46^\circ, 53^\circ, 81^\circ$

8 $52^\circ, 66^\circ, 62^\circ$

9 $52^\circ, 44^\circ, 84^\circ$

10 $105^\circ, 51^\circ, 24^\circ$

11 $50^\circ, 77^\circ, 53^\circ$

12 $79^\circ, 68^\circ, 33^\circ$

13 $99^\circ, 57^\circ, 24^\circ$

14 $119^\circ, 38^\circ, 23^\circ$

15 $44^\circ, 24^\circ, 112^\circ$

16 $33^\circ, 119^\circ, 28^\circ$

17 $23^\circ, 32^\circ, 125^\circ$

18 $85^\circ, 83^\circ, 12^\circ$

19 $51^\circ, 38^\circ, 91^\circ$

20 $73^\circ, 49^\circ, 58^\circ$

21 $23^\circ, 82^\circ, 75^\circ$

22 $39^\circ, 130^\circ, 11^\circ$

23 $35^\circ, 20^\circ, 125^\circ$

24 $71^\circ, 74^\circ, 35^\circ$

ROUND 2 Questions 25–40: classify by sides: equilateral, isosceles or scalene.

25 6 cm, 9 cm, 10 cm

26 4 cm, 4 cm, 4 cm

27 7 cm, 7 cm, 3 cm

28 3 cm, 6 cm, 8 cm

29 7 cm, 7 cm, 6 cm

30 11 cm, 14 cm, 15 cm

31 7 cm, 7 cm, 7 cm

32 8 cm, 8 cm, 4 cm

33 12 cm, 12 cm, 22 cm

34 5 cm, 5 cm, 5 cm

35 12 cm, 12 cm, 12 cm

36 12 cm, 12 cm, 15 cm

37 10 cm, 10 cm, 11 cm

38 12 cm, 12 cm, 4 cm

39 11 cm, 12 cm, 17 cm

40 11 cm, 14 cm, 16 cm

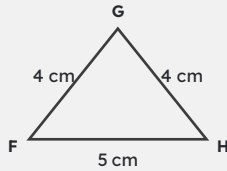
Level 2 · Building up

Medium · 6 questions

WE DO With your teacher: copy what they write on the board.

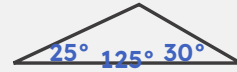
Example 2 Classify the triangle, then find the equal angles.

a Classify $\triangle FGH$. Which angles are equal?



Your teacher works this out. Copy every step.

b Classify by sides and angles.



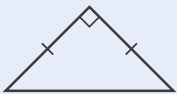
Your teacher works this out. Copy every step.

YOU DO On your own: **Medium** Set C. Classify each triangle by sides and by angles.

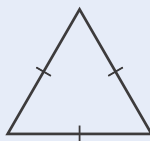
6 QUESTIONS

REMEMBER Angles: **ACUTE** (all $< 90^\circ$), **RIGHT** (one 90°), **OBTUSE** (one $> 90^\circ$).

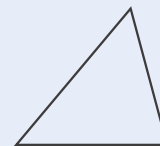
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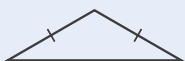
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5



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4



6



SKILL DRILL 2 Medium basics: keep practising

27 questions

YOU DO

On your own: **Medium** Skill drill 2. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 27

Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–15: classify by sides and by angles.

1 angles 52° , 64° , 64°

2 angles 60° , 60° , 60°

3 angles 36° , 72° , 72°

4 angles 90° , 45° , 45°

5 angles 125° , 36° , 19°

6 angles 74° , 53° , 53°

7 angles 121° , 33° , 26°

8 angles 48° , 66° , 66°

9 angles 80° , 50° , 50°

10 angles 106° , 20° , 54°

11 angles 124° , 33° , 23°

12 angles 113° , 20° , 47°

13 angles 122° , 34° , 24°

14 angles 116° , 21° , 43°

15 angles 56° , 62° , 62°

ROUND 2 Questions 16–27: is it possible to draw this triangle? Write yes or no.

16 an acute scalene triangle

17 a triangle with sides 5 cm, 5 cm and 9 cm

18 an obtuse equilateral triangle

19 a triangle with two right angles

20 a triangle with two obtuse angles

21 an acute isosceles triangle

22 a triangle with angles 100° , 40° , 40°

23 a triangle with angles 30° , 60° , 90°

24 a right-angled isosceles triangle

25 a triangle with angles 90° , 60° and 40°

26 a triangle with sides 3 cm, 4 cm and 10 cm

27 an obtuse isosceles triangle

Level 3 · Challenge

Challenging · 4 questions

WE DO With your teacher: copy what they write on the board.

Example 3 Classify each triangle from its description.

a My angles are 60° , 80° and 40° . Classify me.

Your teacher works this out. Copy every step.

b All my angles are equal. Classify me.

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Classify each described triangle by sides and by angles.

4 QUESTIONS

REMEMBER Equal sides have matching tick marks.

1 My angles are 60° , 80° and 40° .

3 My angles are 30° , 30° and 120° .

2 My sides are 5 cm, 5 cm and 7 cm, and one angle is 90° . Is this possible? Explain.

4 Can a triangle be right-angled and equilateral? Explain.



Extension · Stretch yourself

Finished Set D? Stretch yourself here.



Worked example

An isosceles triangle has one angle of 40° . What could the other two angles be?

1. If 40° is one of the equal angles: $40^\circ, 40^\circ, 100^\circ$
2. If 40° is the apex: $(180 - 40) \div 2 = 70^\circ$

Answer: 40° and 100° , or 70° and 70°

YOU DO

On your own: read the worked example, then try these.

- E1** An isosceles triangle has one angle of 100° . Find the other two angles.

- E2** An isosceles triangle has one angle of 60° . What type of triangle is it?

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO

Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Classify a triangle with sides 6, 6 and 6 cm.

Level 2

Classify a triangle with angles 90° , 45° and 45° .

Level 3

Classify a triangle with angles 110° , 35° and 35° by sides and angles.

Lesson summary

Missed the lesson? Start here.

- Remember**
1. Sides: **equilateral** (3 equal), **isosceles** (2 equal), **scalene** (none).
 2. Angles: **acute** (all $< 90^\circ$), **right** (one 90°), **obtuse** (one $> 90^\circ$).
 3. Equal sides have matching tick marks.
 4. Angles opposite equal sides are equal.

Level 1

3 equal sides

1. All angles 60°

Answer: equilateral, acute

Level 2

$\triangle FGH$: 4 cm, 4 cm, 5 cm

1. Two equal sides

Answer: isosceles; $\angle F = \angle H$

Level 3

$25^\circ, 30^\circ, 125^\circ$

1. One obtuse angle

Answer: scalene, obtuse

Extension

Isosceles with a 40° angle

1. Two cases

Answer: $40^\circ, 100^\circ$ or $70^\circ, 70^\circ$

My reflection: what did I learn today? Which question was hardest, and why?

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
6.06

Angle sum of a triangle

Date _____

Learning intentions

We are learning to:

- use the angle sum of a triangle to find unknown angles

Success criteria

I can:

- ☐ know the angles in a triangle add to 180°
- ☐ use equal angles in isosceles and equilateral triangles
- ☐ give reasons for each step

Key terms Write the definition as your teacher explains it.

Angle sum

Reason

Notes

Level 1 · Getting started

Easy · 15 questions

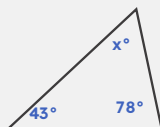
WE DO

With your teacher: copy what they write on the board.

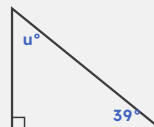
Example 1

Find the unknown angle. Angles in a triangle add to 180° .

a Find x .



b Find u .



Your teacher works this out. Copy every step.

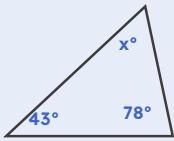
Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Find the value of each variable.

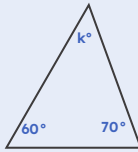
6 QUESTIONS

REMEMBER Angles in a triangle add to 180° .

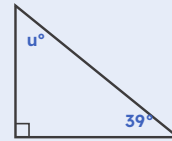
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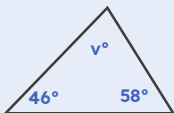
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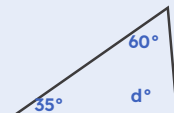
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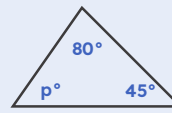
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YOU DO On your own: **Easy** Set A2. Quick drill: two angles of a triangle are given. Find the third. 6 QUESTIONS

1 50° and 60°

2 90° and 35°

3 45° and 45°

4 100° and 30°

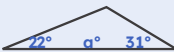
5 72° and 72°

6 25° and 115°

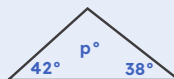
YOU DO On your own: **Easy** Set B. Find the value of each variable.

3 QUESTIONS

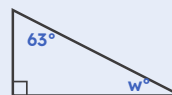
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SKILL DRILL 1 Easy basics: practise until it is automatic

40 questions

YOU DO

On your own: **Easy** Skill drill 1. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 40

Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–24: two angles of a triangle are given. Find the third. (Angles add to 180° .)1 68° and 96°

2 85° and 20°

3 43° and 53°

4 46° and 72°

5 29° and 22°

6 82° and 73°

7 15° and 32°

8 112° and 31°

9 85° and 53°

10 37° and 64°

11 65° and 76°

12 24° and 103°

13 63° and 26°

14 37° and 95°

15 47° and 32°

16 116° and 39°

17 86° and 68°

18 75° and 50°

19 93° and 64°

20 105° and 50°

21 91° and 67°

22 52° and 51°

23 88° and 62°

24 56° and 49°

ROUND 2 Questions 25–40: a right-angled triangle has this angle. Find the other one.25 11°

26 18°

27 78°

28 74°

29 38°

30 39°

31 10°

32 64°

33 61°

34 57°

35 75°

36 47°

37 28°

38 67°

39 23°

40 48°

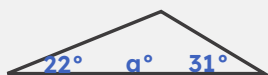
Level 2 · Building up

Medium · 6 questions

WE DO With your teacher: copy what they write on the board.

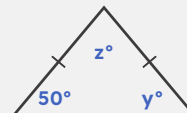
Example 2 Find the unknown angles.

a Find a .



Your teacher works this out. Copy every step.

b Find y and z .



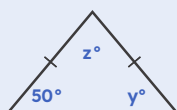
Your teacher works this out. Copy every step.

YOU DO On your own: **Medium** Set C. Isosceles and equilateral triangles. Find each variable.

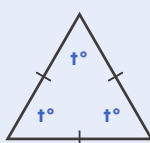
6 QUESTIONS

REMEMBER Add the known angles, subtract from 180° .

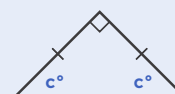
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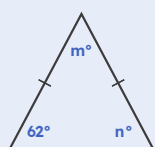
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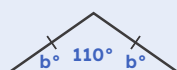
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6



SKILL DRILL 2 Medium basics: keep practising

28 questions

YOU DO

On your own: **Medium** Skill drill 2. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 28 Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–16: isosceles triangle: find the missing angles.

1 apex angle 24°

2 base angle 75°

3 base angle 55°

4 apex angle 50°

5 base angle 25°

6 apex angle 150°

7 base angle 20°

8 base angle 31°

9 base angle 39°

10 base angle 54°

11 base angle 21°

12 base angle 72°

13 base angle 38°

14 base angle 79°

15 base angle 46°

16 apex angle 82°

ROUND 2 Questions 17–28: the angles of a triangle are given. Find x .

17 x° , $2x + 15^\circ$, 102°

18 $2x + 20^\circ$, x° , 16°

19 x° , $x + 20^\circ$, 90°

20 $2x + 15^\circ$, x° , 117°

21 x° , $x + 15^\circ$, 71°

22 $x + 20^\circ$, $2x^\circ$, 19°

23 $2x + 10^\circ$, $2x + 15^\circ$, 47°

24 x° , $3x^\circ$, 120°

25 x° , $2x + 20^\circ$, 85°

26 $x + 10^\circ$, x° , 42°

27 $3x^\circ$, x° , 48°

28 $x + 15^\circ$, $x + 5^\circ$, 122°

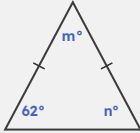
Level 3 · Challenge

Challenging · 4 questions

WE DO With your teacher: copy what they write on the board.

Example 3 Find the unknown angles in the isosceles triangles.

a Find n and m (isosceles).



Your teacher works this out. Copy every step.

b Find $\angle PQR$.



Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Find the unknowns. Give reasons.

4 QUESTIONS

REMEMBER Isosceles: base angles are equal.

1 The angles of a triangle are x° , $2x^\circ$ and $3x^\circ$. Find x .

3 Can a triangle have angles of 100° and 90° ? Explain.

2 An isosceles triangle has an apex angle of 36° . Find the base angles.

4 Two angles of a triangle are equal and the third is 20° more than each of them. Find all three.



Extension · Stretch yourself

Finished Set D? Stretch yourself here.



Worked example

Prove that the angles of a triangle add to 180° using a line parallel to the base.

1. Draw a line through the top vertex parallel to the base
2. The two outer angles equal the base angles (alternate angles)
3. The three angles at the top form a straight line: 180°

Answer: $a + b + c = 180^\circ$

YOU DO

On your own: read the worked example, then try these.

- E1 Tear the corners off a paper triangle and place them together. What do you see?

- E2 Use the angle sum to explain why a triangle can have only one obtuse angle.

YOU DO

Show off your skill

Choose ONE question. Show your working.

Name _____ Class _____

Level 1

Two angles of a triangle are 65° and 45° . Find the third.

Level 2

A right-angled triangle has an angle of 28° . Find the other angle.

Level 3

An isosceles triangle has base angles of 67° . Find the apex angle.

Lesson summary

Missed the lesson? Start here.

Remember

- Angles in a triangle add to **180°**.
- Isosceles: base angles are equal.

- Add the known angles, subtract from 180°.
- Equilateral: every angle is 60°.

Level 1

43°, 78°, x°

1. $180 - 43 - 78$

Answer: $x = 59$

Level 2

Right angle and 39°

1. $180 - 90 - 39$

Answer: $u = 51$

Level 3

Isosceles, apex 72° at P? no: base 72°

1. $\angle PQR = 180 - 72 - 72$

Answer: 36°

Extension

Proof

- Parallel line and alternate angles

Answer: 180°

My reflection: what did I learn today? Which question was hardest, and why?

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
6.07

Exterior angle of a triangle

Date _____

Learning intentions

We are learning to:

- use the exterior angle property of a triangle

Success criteria

I can:

- ☐ identify an **exterior angle** of a triangle
- ☐ know the exterior angle equals the sum of the two interior opposite angles
- ☐ find unknown angles using this rule

Key terms Write the definition as your teacher explains it.

Exterior angle

Interior opposite angles

Notes

Level 1 · Getting started

Easy · 15 questions

WE DO

With your teacher: copy what they write on the board.

Example 1

Find the exterior angle.

a Find x .



b Find x .



Your teacher works this out. Copy every step.

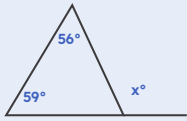
Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Find the value of each variable.

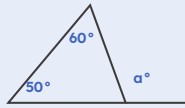
6 QUESTIONS

REMEMBER An **EXTERIOR ANGLE** is made by extending one side.

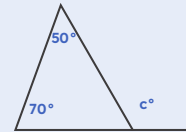
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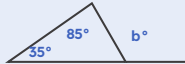
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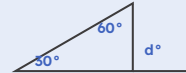
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YOU DO On your own: **Easy** Set A2. Quick drill: interior opposite angles are given. Find the exterior angle.

6 QUESTIONS

1

40° and 70°

2

55° and 65°

3

30° and 90°

4

62° and 58°

5

25° and 45°

6

80° and 35°

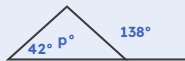
YOU DO On your own: **Easy** Set B. Find the value of each variable.

3 QUESTIONS

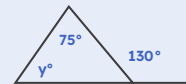
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SKILL DRILL 1 Easy basics: practise until it is automatic

40 questions

YOU DO

On your own: **Easy** Skill drill 1. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 40

Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–24: the two interior opposite angles are given. Find the exterior angle (add them).

1 74° and 20°

2 90° and 34°

3 49° and 38°

4 17° and 122°

5 30° and 64°

6 32° and 40°

7 25° and 63°

8 46° and 43°

9 25° and 55°

10 72° and 67°

11 68° and 36°

12 95° and 39°

13 94° and 49°

14 36° and 97°

15 24° and 85°

16 64° and 71°

17 92° and 62°

18 97° and 52°

19 83° and 37°

20 20° and 21°

21 33° and 31°

22 92° and 21°

23 60° and 87°

24 74° and 82°

ROUND 2 Questions 25–40: an exterior angle is given. Find the interior angle next to it.

25 143°

26 146°

27 102°

28 75°

29 118°

30 78°

31 49°

32 93°

33 128°

34 66°

35 55°

36 124°

37 20°

38 41°

39 22°

40 44°

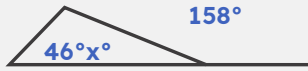
Level 2 · Building up

Medium · 6 questions

WE DO With your teacher: copy what they write on the board.

Example 2 Find the unknown interior angle.

a Find x .



b Find p .



Your teacher works this out. Copy every step.

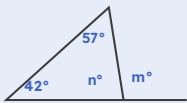
Your teacher works this out. Copy every step.

YOU DO On your own: **Medium** Set C. Find each variable. Give reasons.

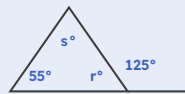
6 QUESTIONS

REMEMBER Exterior angle = sum of the two **INTERIOR OPPOSITE** angles.

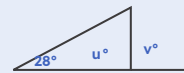
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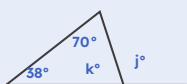
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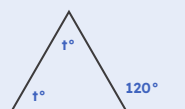
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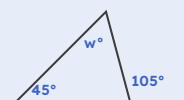
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SKILL DRILL 2 Medium basics: keep practising

28 questions

YOU DO

On your own: **Medium** Skill drill 2. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 28 Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1

Questions 1–16: the exterior angle and one interior opposite angle are given. Find the other interior opposite angle.

1 exterior 104° , 23°

2 exterior 125° , 32°

3 exterior 137° , 45°

4 exterior 76° , 43°

5 exterior 70° , 45°

6 exterior 107° , 90°

7 exterior 168° , 112°

8 exterior 122° , 68°

9 exterior 113° , 102°

10 exterior 54° , 41°

11 exterior 161° , 18°

12 exterior 102° , 18°

13 exterior 92° , 16°

14 exterior 169° , 39°

15 exterior 163° , 81°

16 exterior 55° , 30°

ROUND 2

Questions 17–28: find x . The exterior angle equals the sum of the interior opposite angles.

17 exterior 83° , interior opposite 66° and x°

18 exterior 133° , interior opposite 54° and x°

19 exterior 104° , interior opposite 47° and x°

20 exterior 101° , interior opposite 65° and x°

21 exterior 79° , interior opposite 21° and x°

22 exterior 110° , interior opposite 40° and x°

23 exterior 136° , interior opposite 70° and x°

24 exterior 45° , interior opposite 20° and x°

25 exterior 72° , interior opposite 35° and x°

26 exterior 42° , interior opposite 22° and x°

27 exterior 67° , interior opposite 24° and x°

28 exterior 97° , interior opposite 22° and x°

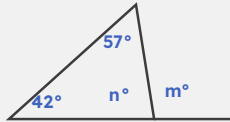
Level 3 · Challenge

Challenging · 4 questions

WE DO With your teacher: copy what they write on the board.

Example 3 Find both unknowns, then explain.

a Find m and n .



Your teacher works this out. Copy every step.

b Why is the exterior angle always larger than each interior opposite angle?

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Harder problems.

4 QUESTIONS

REMEMBER Exterior + adjacent interior = 180° .

1 The interior opposite angles are x° and $2x^\circ$. The exterior angle is 111° . Find x .

3 What do the three exterior angles of a triangle (one at each vertex) add to? Test with an equilateral triangle.

2 The exterior angle of an isosceles triangle at the base is 110° . Find all three interior angles.

4 Can an exterior angle be acute? Explain.

Extension · Stretch yourself

Finished Set D? Stretch yourself here.

Worked example

Prove the exterior angle property using the angle sum.

1. $a + b + c = 180$ (angle sum)
2. $c + x = 180$ (straight line)
3. So $x = a + b$

Answer: The exterior angle equals the sum of the interior opposite angles

YOU DO

On your own: read the worked example, then try these.

- E1 An exterior angle is 4 times the interior angle next to it. Find the exterior angle.

- E2 The exterior angle is 130° and one interior opposite angle is 20° more than the other. Find both.

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO

Show off your skill

Choose ONE question. Show your working.

Name _____ Class _____

Level 1

Interior opposite angles 40° and 75° . Find the exterior angle.

Level 2

The exterior angle is 125° and one interior opposite angle is 60° . Find the other.

Level 3

The exterior angle is 140° . Find the adjacent interior angle.

Lesson summary

Missed the lesson? Start here.

- Remember**
1. An **exterior angle** is made by extending one side.
 2. Exterior angle = sum of the two **interior opposite** angles.
 3. Exterior + adjacent interior = 180° .
 4. Give the reason: "exterior angle of a triangle".

Level 1

59° and 56° opposite

1. $59 + 56$

Answer: $x = 115$

Level 2

$46^\circ + x = 158^\circ$

1. $158 - 46$

Answer: $x = 112$

Level 3

$42^\circ + p = 138^\circ$

1. $138 - 42$

Answer: $p = 96$

Extension

Proof

1. Angle sum and straight line

Answer: $x = a + b$

My reflection: what did I learn today? Which question was hardest, and why?

This space is left blank on purpose. It is the back of your exit ticket.

LESSON 6.08

Quadrilaterals

Date _____

Learning intentions

We are learning to:

- classify quadrilaterals
- describe the symmetry and definitions of special quadrilaterals

Success criteria

I can:

- ☐ decide if a quadrilateral is **convex** or **non-convex**
- ☐ name the special quadrilaterals
- ☐ give the definition of each special quadrilateral

Key terms Write the definition as your teacher explains it.

Convex

Parallelogram

Rhombus

Notes

Level 1 · Getting started

Easy · 15 questions

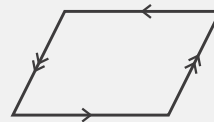
WE DO With your teacher: copy what they write on the board.

Example 1 Convex or non-convex? Name the quadrilateral.

a Convex or non-convex?



b Name this quadrilateral.



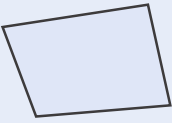
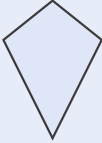
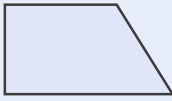
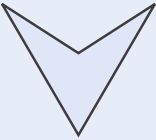
Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Convex or non-convex?

6 QUESTIONS

REMEMBER **TRAPEZIUM**: at least one pair of parallel sides. **PARALLELOGRAM**: both pairs parallel.

1 	3 	5 
2 	4 	6 

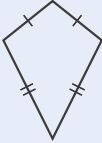
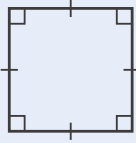
YOU DO On your own: **Easy** Set A2. Quick drill: name the quadrilateral.

6 QUESTIONS

- | | |
|--|---|
| <p>1 4 equal sides, 4 right angles</p> <p>_____</p> | <p>2 both pairs of opposite sides parallel</p> <p>_____</p> |
| <p>3 one pair of parallel sides</p> <p>_____</p> | <p>4 parallelogram with 4 equal sides</p> <p>_____</p> |
| <p>5 parallelogram with a right angle</p> <p>_____</p> | <p>6 two pairs of equal adjacent sides</p> <p>_____</p> |

YOU DO On your own: **Easy** Set B. Name each special quadrilateral.

3 QUESTIONS

1 	2 	3 
--	--	--

SKILL DRILL 1 Easy basics: practise until it is automatic

22 questions

YOU DO

On your own: **Easy** Skill drill 1. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 22

Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–12: name the quadrilateral.

1 2 pairs of equal adjacent sides

2 a parallelogram with 4 equal sides

3 a parallelogram with a right angle

4 a trapezium with equal legs

5 2 pairs of parallel sides

6 4 equal sides and 4 right angles

7 a rectangle with 4 equal sides

8 diagonals cross at 90° and one diagonal bisects the other

9 opposite angles equal, no right angles, sides not all equal

10 4 equal sides, no right angles

11 exactly 1 pair of parallel sides

12 4 right angles, opposite sides equal

ROUND 2 Questions 13–22: true or false?

13 Every rectangle is a parallelogram.

14 Every square is a rectangle.

15 Every rhombus is a parallelogram.

16 A quadrilateral has 4 sides.

17 A trapezium has two pairs of parallel sides.

18 Every parallelogram is a rectangle.

19 A rhombus has 4 right angles.

20 Every rectangle is a square.

21 Every square is a rhombus.

22 A kite is a parallelogram.

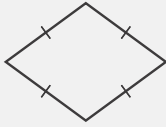
Level 2 · Building up

Medium · 6 questions

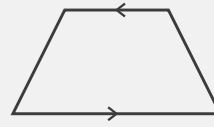
WE DO With your teacher: copy what they write on the board.

Example 2 Name each quadrilateral.

a Name this quadrilateral.



b Name this quadrilateral.



Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** Set C. For each shape in Set B and Examples 2–4, write the number of axes of symmetry and the order of rotational symmetry.

6 QUESTIONS

REMEMBER RHOMBUS: parallelogram with all sides equal. **RECTANGLE:** parallelogram with a right angle.

1 Rectangle

3 Square

5 Rhombus

2 Kite

4 Parallelogram

6 Trapezium (isosceles)

SKILL DRILL 2 Medium basics: keep practising

15 questions

YOU DO

On your own: **Medium** Skill drill 2. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 15 Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–7: how many pairs of parallel sides? How many right angles?

1 isosceles trapezium

2 square

3 rectangle

4 parallelogram

5 trapezium

6 kite

7 rhombus

ROUND 2 Questions 8–15: name the most specific quadrilateral.

8 a quadrilateral with only one diagonal as an axis of symmetry

9 a quadrilateral with 2 pairs of parallel sides and equal diagonals

10 a quadrilateral whose diagonals bisect each other

11 a quadrilateral whose diagonals are equal and bisect each other at 90°

12 a quadrilateral whose diagonals bisect each other at right angles

13 a kite with 4 equal sides

14 a quadrilateral with one pair of parallel sides and equal legs

15 a rhombus with a right angle



Level 3 · Challenge

Challenging · 4 questions

WE DO

With your teacher: copy what they write on the board.

Example 3

Give the definitions.

a What is the definition of a rhombus?

Your teacher works this out. Copy every step.

b What am I? I have one pair of parallel sides.

Your teacher works this out. Copy every step.

YOU DO

On your own: **Challenging** Set D. Definitions.

4 QUESTIONS

REMEMBER **SQUARE**: rectangle with all sides equal. **KITE**: two pairs of equal adjacent sides.

1 What quadrilateral am I? I have both pairs of opposite sides parallel.

3 Is a square a rhombus? Explain using definitions.

2 Write the definition of a rectangle.

4 Is every rectangle a parallelogram? Explain.

Extension · Stretch yourself

Finished Set D? Stretch yourself here.

Worked example

Draw a family tree of the quadrilaterals.

1. Quadrilateral → trapezium → parallelogram
2. Parallelogram → rectangle and rhombus
3. Rectangle and rhombus → square

Answer: A square is a special rectangle and a special rhombus

YOU DO

On your own: read the worked example, then try these.

- E1 Where does a kite fit in the family tree? Is a rhombus a kite?

- E2 Which special quadrilaterals have exactly 2 axes of symmetry?

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO Show off your skill

Choose ONE question. Show your working.

Name _____ Class _____

Level 1

Is a square convex?

Level 2

Name a quadrilateral with exactly one pair of parallel sides.

Level 3

What is the definition of a parallelogram?

Lesson summary

Missed the lesson? Start here.

- Definitions**
1. **Trapezium:** at least one pair of parallel sides.
Parallelogram: both pairs parallel.
 2. **Rhombus:** parallelogram with all sides equal.
Rectangle: parallelogram with a right angle.
 3. **Square:** rectangle with all sides equal. **Kite:** two pairs of equal adjacent sides.
 4. **Convex:** all angles less than 180° .

Level 1

Dart shape

1. One angle is reflex

Answer: non-convex

Level 2

Opposite sides parallel

1. Both pairs

Answer: parallelogram

Level 3

Rhombus definition

1. Parallelogram, all sides equal

Answer: a parallelogram with 4 equal sides

Extension

Square in the family

1. Rectangle and rhombus

Answer: both

My reflection: what did I learn today? Which question was hardest, and why?

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
6.09

Angle sum of a quadrilateral

Date _____

Learning intentions

We are learning to:

- use the angle sum of a quadrilateral to find unknown angles

Success criteria

I can:

- ☐ know the angles in a quadrilateral add to 360°
- ☐ find an unknown angle
- ☐ use right angles and equal angles in special quadrilaterals

Key terms Write the definition as your teacher explains it.

Angle sum

Quadrilateral

Notes

Level 1 · Getting started

Easy · 15 questions

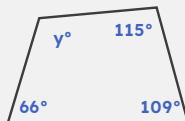
WE DO

With your teacher: copy what they write on the board.

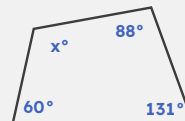
Example 1

Find the unknown angle. Angles in a quadrilateral add to 360° .

a Find y .



b Find x .



Your teacher works this out. Copy every step.

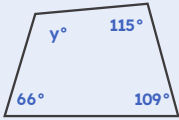
Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Find the value of each variable.

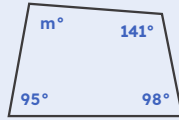
6 QUESTIONS

REMEMBER Angles in a quadrilateral add to 360° .

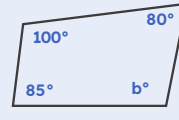
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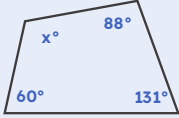
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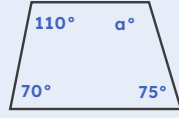
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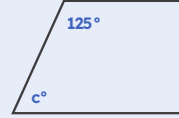
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4



6



YOU DO On your own: **Easy** Set A2. Quick drill: three angles of a quadrilateral are given. Find the fourth.

6 QUESTIONS

1

$90^\circ, 90^\circ, 90^\circ$

2

$100^\circ, 80^\circ, 100^\circ$

3

$70^\circ, 110^\circ, 95^\circ$

4

$120^\circ, 60^\circ, 60^\circ$

5

$85^\circ, 85^\circ, 100^\circ$

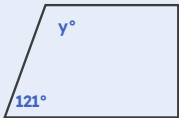
6

$45^\circ, 135^\circ, 90^\circ$

YOU DO On your own: **Easy** Set B. Find each variable.

3 QUESTIONS

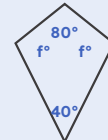
1



2



3



SKILL DRILL 1 Easy basics: practise until it is automatic

36 questions

YOU DO

On your own: **Easy** Skill drill 1. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 36

Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–24: three angles of a quadrilateral are given. Find the fourth. (Angles add to 360° .)

1 $126^\circ, 108^\circ, 72^\circ$

2 $104^\circ, 139^\circ, 77^\circ$

3 $93^\circ, 100^\circ, 91^\circ$

4 $139^\circ, 112^\circ, 50^\circ$

5 $96^\circ, 64^\circ, 94^\circ$

6 $97^\circ, 135^\circ, 65^\circ$

7 $124^\circ, 104^\circ, 52^\circ$

8 $118^\circ, 106^\circ, 80^\circ$

9 $71^\circ, 57^\circ, 88^\circ$

10 $86^\circ, 108^\circ, 53^\circ$

11 $131^\circ, 58^\circ, 115^\circ$

12 $108^\circ, 87^\circ, 131^\circ$

13 $76^\circ, 125^\circ, 75^\circ$

14 $125^\circ, 78^\circ, 76^\circ$

15 $103^\circ, 122^\circ, 80^\circ$

16 $93^\circ, 121^\circ, 60^\circ$

17 $55^\circ, 80^\circ, 81^\circ$

18 $139^\circ, 70^\circ, 92^\circ$

19 $129^\circ, 92^\circ, 51^\circ$

20 $115^\circ, 62^\circ, 76^\circ$

21 $111^\circ, 94^\circ, 83^\circ$

22 $74^\circ, 69^\circ, 75^\circ$

23 $95^\circ, 89^\circ, 122^\circ$

24 $106^\circ, 132^\circ, 59^\circ$

ROUND 2 Questions 25–36: a parallelogram has this angle. Find the other three angles.

25 111°

26 26°

27 112°

28 42°

29 119°

30 23°

31 31°

32 146°

33 27°

34 36°

35 106°

36 160°

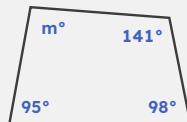
Level 2 · Building up

Medium · 6 questions

WE DO With your teacher: copy what they write on the board.

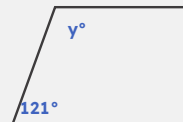
Example 2 Find the unknown angle.

a Find m .



Your teacher works this out. Copy every step.

b Find y .



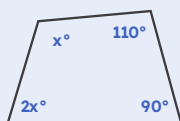
Your teacher works this out. Copy every step.

YOU DO On your own: **Medium** Set C. Find each variable. Give reasons.

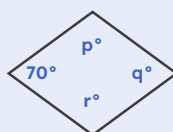
6 QUESTIONS

REMEMBER A diagonal makes 2 triangles: $2 \times 180^\circ$.

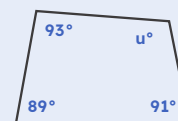
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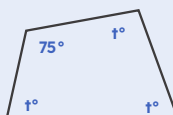
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2



4



6



SKILL DRILL 2 Medium basics: keep practising

24 questions

YOU DO

On your own: **Medium** Skill drill 2. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 24 Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–12: a kite has two equal opposite angles, x . Find x .1 other angles 82° and 84°

2 other angles 52° and 52°

3 other angles 96° and 58°

4 other angles 77° and 65°

5 other angles 44° and 146°

6 other angles 116° and 90°

7 other angles 94° and 134°

8 other angles 85° and 123°

9 other angles 63° and 97°

10 other angles 43° and 59°

11 other angles 58° and 74°

12 other angles 62° and 106°

ROUND 2 Questions 13–24: the angles of a quadrilateral are given. Find x .13 $x^\circ, 3x^\circ, 3x^\circ, 10^\circ$

14 $2x + 10^\circ, x + 15^\circ, 3x^\circ, 191^\circ$

15 $2x^\circ, x + 10^\circ, 2x + 10^\circ, 75^\circ$

16 $x + 20^\circ, 2x^\circ, 3x^\circ, 280^\circ$

17 $x^\circ, 3x^\circ, x^\circ, 105^\circ$

18 $x^\circ, 3x^\circ, 3x^\circ, 199^\circ$

19 $2x^\circ, 2x^\circ, x^\circ, 285^\circ$

20 $x + 5^\circ, 2x + 5^\circ, 2x^\circ, 215^\circ$

21 $2x^\circ, 2x^\circ, 3x^\circ, 157^\circ$

22 $2x + 15^\circ, 3x^\circ, x^\circ, 105^\circ$

23 $x^\circ, x^\circ, x^\circ, 252^\circ$

24 $2x + 10^\circ, x + 15^\circ, x^\circ, 19^\circ$



Level 3 · Challenge

Challenging · 4 questions

WE DO With your teacher: copy what they write on the board.**Example 3** Explain, then find the angles.

- a** Why do the angles in a quadrilateral add to 360° ?

Your teacher works this out. Copy every step.

- b** Three angles of a quadrilateral are equal and the fourth is 90° . Find the equal angles.

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Think it through.

4 QUESTIONS

REMEMBER Add the known angles, subtract from 360° .

- 1** The angles of a quadrilateral are x° , $2x^\circ$, $3x^\circ$ and $4x^\circ$. Find each angle.

- 3** A parallelogram has one angle of 58° . Find the other three.

- 2** Can a quadrilateral have 4 obtuse angles? Explain.

- 4** A kite has angles 120° and 60° between the unequal sides. Find the other two (equal) angles.

Extension · Stretch yourself

Finished Set D? Stretch yourself here.

Worked example

Find the angle sum of a pentagon by splitting it into triangles.

1. Draw diagonals from one vertex: 3 triangles
2. $3 \times 180^\circ$

Answer: 540°

YOU DO

On your own: read the worked example, then try these.

- E1 Find the angle sum of a hexagon.

- E2 Find each angle of a regular pentagon.

YOU DO

Show off your skill

Choose ONE question. Show your working.

Name _____ Class _____

Level 1

Three angles are 90° , 90° and 70° . Find the fourth.

Level 2

Three angles are 100° , 85° and 95° . Find the fourth.

Level 3

A parallelogram has an angle of 65° . Find the other angles.

Lesson summary

Missed the lesson? Start here.

- Remember**
- Angles in a quadrilateral add to 360° .
 - A diagonal makes 2 triangles: $2 \times 180^\circ$.
 - Add the known angles, subtract from 360° .
 - Use properties: opposite angles in a parallelogram are equal.

Level 1

$66^\circ, 109^\circ, 115^\circ, y^\circ$

1. $360 - 290$

Answer: $y = 70$

Level 2

$60^\circ, 131^\circ, 88^\circ, x^\circ$

1. $360 - 279$

Answer: $x = 81$

Level 3

Two right angles and 121°

1. $360 - 90 - 90 - 121$

Answer: $y = 59$

Extension

Pentagon

1. 3 triangles

Answer: 540°

My reflection: what did I learn today? Which question was hardest, and why?

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
6.10

Properties of quadrilaterals

Date _____

Learning intentions

We are learning to:

- describe the side, angle and diagonal properties of special quadrilaterals

Success criteria

I can:

- ☐ list the properties of a parallelogram
- ☐ describe how the diagonals behave in each special quadrilateral
- ☐ identify a quadrilateral from its properties

Key terms Write the definition as your teacher explains it.

Diagonal

Bisect

Notes

Level 1 · Getting started

Easy · 18 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Use the properties of a parallelogram.

a List 2 properties of a parallelogram.

b EFGH is a parallelogram. Which is NOT true? A $\angle EFG = \angle EHG$ B $EF = HG$ C $EF = FG$ D $\angle EFG + \angle FGH = 180^\circ$

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO

On your own: **Easy** Set A. Questions 1–6: True or false?

12 QUESTIONS

REMEMBER Parallelogram: diagonals **BISECT** each other.

1 A rectangle has equal diagonals.

2 A rhombus has 4 right angles.

3 The diagonals of a square are perpendicular.

4 A kite has 2 pairs of parallel sides.

5 A parallelogram has opposite sides equal.

6 A trapezium always has equal diagonals.

Questions 7–12: true or false?

7 A square is a rectangle.

8 A rectangle is a square.

9 A rhombus is a parallelogram.

10 A kite has parallel sides.

11 A square is a rhombus.

12 A trapezium is a parallelogram.

YOU DO

On your own: **Easy** Set B. Name all the quadrilaterals that have:

6 QUESTIONS

1 4 equal sides

4 diagonals perpendicular

2 4 right angles

5 exactly one pair of parallel sides

3 opposite angles equal

6 diagonals that bisect the angles

SKILL DRILL 1 Easy basics: practise until it is automatic

22 questions

YOU DO On your own: **Easy** Skill drill 1. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 22 Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–16: true or false?

- | | |
|--|---|
| 1 The diagonals of a rhombus bisect its angles.
_____ | 2 Co-interior angles in a parallelogram add to 180° .
_____ |
| 3 The opposite sides of a parallelogram are equal.
_____ | 4 A rhombus has equal diagonals.
_____ |
| 5 The adjacent sides of a rectangle are always equal.
_____ | 6 A square is a special kite.
_____ |
| 7 The opposite angles of a rhombus are equal.
_____ | 8 A kite has two pairs of equal opposite sides.
_____ |
| 9 The diagonals of a rhombus meet at 90° .
_____ | 10 A trapezium always has a right angle.
_____ |
| 11 The diagonals of a square are equal and meet at 90° .
_____ | 12 The diagonals of a parallelogram are equal.
_____ |
| 13 The diagonals of a kite meet at 90° .
_____ | 14 The diagonals of a parallelogram bisect each other.
_____ |
| 15 The diagonals of a rectangle are equal.
_____ | 16 All the angles of a rectangle are 90° .
_____ |

ROUND 2 Questions 17–22: which quadrilaterals have this property? Name all you know.

- | | |
|--|--|
| 17 diagonals meet at 90°
_____ | 18 diagonals bisect each other
_____ |
| 19 all sides equal
_____ | 20 diagonals are equal
_____ |
| 21 opposite angles equal
_____ | 22 exactly one pair of parallel sides
_____ |

Level 2 · Building up

Medium · 6 questions

WE DO With your teacher: copy what they write on the board.

Example 2 Name all the quadrilaterals that fit.

a What am I? All my angles are equal.

Your teacher works this out. Copy every step.

b What am I? My diagonals are equal and bisect each other.

Your teacher works this out. Copy every step.

YOU DO On your own: **Medium** Set C. Draw the diagonals and describe them (equal? bisect? perpendicular?).

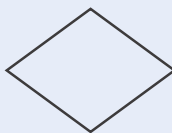
6 QUESTIONS

REMEMBER Rectangle: diagonals equal and bisect.

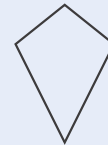
1 Parallelogram



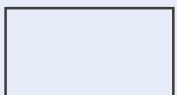
3 Rhombus



5 Kite



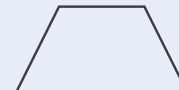
2 Rectangle



4 Square



6 Trapezium



SKILL DRILL 2 Medium basics: keep practising

18 questions

YOU DO

On your own: **Medium** Skill drill 2. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 18 Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–12: use the properties to find the missing value.

- | | |
|---|--|
| 1 rhombus: one angle is 50° . The opposite angle?
_____ | 2 parallelogram: sides 9 cm and 5 cm. The perimeter?
_____ |
| 3 kite: angles 100° , 80° , 80° . The fourth?
_____ | 4 isosceles trapezium: one base angle is 65° . The other base angle?
_____ |
| 5 rectangle: one diagonal is 13 cm. The other?
_____ | 6 parallelogram: one angle is 70° . The angle next to it?
_____ |
| 7 rectangle: half a diagonal is 6 cm. The whole diagonal?
_____ | 8 square: diagonals meet at what angle?
_____ |
| 9 parallelogram: one angle is 125° . The opposite angle?
_____ | 10 rhombus: one side is 7 cm. The perimeter?
_____ |
| 11 square: perimeter 48 cm. The side?
_____ | 12 rhombus: perimeter 36 cm. One side?
_____ |

ROUND 2 Questions 13–18: give a reason: why is each statement true?

- | | |
|---|---|
| 13 A square is a rhombus.
_____ | 14 A rectangle is a parallelogram.
_____ |
| 15 A rhombus is a parallelogram.
_____ | 16 A square is a kite.
_____ |
| 17 A square is a rectangle.
_____ | 18 A rhombus is a kite.
_____ |



Level 3 · Challenge

Challenging · 4 questions

WE DO With your teacher: copy what they write on the board.**Example 3** Name all the quadrilaterals that fit.**a** What am I? My diagonals bisect each other.

Your teacher works this out. Copy every step.

b What am I? My opposite angles are equal.

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Use properties.

4 QUESTIONS

REMEMBER Rhombus: diagonals bisect at 90° and bisect the angles.**1** ABCD is a parallelogram with $AB = 7$ cm, $BC = 4$ cm and $\angle A = 65^\circ$. Find CD, AD and $\angle C$.**3** What am I? My diagonals are equal, bisect each other and are perpendicular.**2** The diagonals of a rhombus are 6 cm and 8 cm. What angle do they meet at?**4** What am I? I have one axis of symmetry and diagonals that are perpendicular.



Extension · Stretch yourself

Finished Set D? Stretch yourself here.



Worked example

A rhombus has diagonals 10 cm and 24 cm.
Find its side length.

1. Diagonals bisect each other at right angles:
halves are 5 and 12
2. $\text{Side}^2 = 5^2 + 12^2 = 169$ (Pythagoras, from Stage 5)

Answer: 13 cm

YOU DO

On your own: read the worked example,
then try these.

- E1** A rectangle has sides 6 cm and 8 cm. Its diagonals are 10 cm. How far is the centre from each corner?

- E2** Explain why a square's diagonals make angles of 45° with its sides.

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO

Show off your skill

Choose **ONE** question. Show
your working.

Name _____ Class _____

Level 1

Name a quadrilateral with equal diagonals.

Level 2

State one property of the diagonals of a rhombus.

Level 3

What am I? My opposite sides are equal and all angles are 90° .

Lesson summary

Missed the lesson? Start here.

- Diagonals**
1. Parallelogram: diagonals **bisect** each other.
 2. Rectangle: diagonals equal and bisect.
 3. Rhombus: diagonals bisect at 90° and bisect the angles.
 4. Square: all of these. Kite: one diagonal bisects the other at 90° .

Level 1

Parallelogram properties

1. Opposite sides equal
2. Opposite angles equal

Answer: any 2

Level 2

EFGH: which is not true?

1. Adjacent sides need not be equal

Answer: C

Level 3

Diagonals equal and bisect

1. Rectangle family

Answer: rectangle, square

Extension

Rhombus diagonals 10, 24

1. Halves 5, 12

Answer: 13 cm

My reflection: what did I learn today? Which question was hardest, and why?

This space is left blank on purpose. It is the back of your exit ticket.